MECHANICAL SPECIFICATIONS

PART 1- GENERAL

1.01 GENERAL

A. THE LATEST EDITION OF AIA DOCUMENTS A201 GENERAL CONDITIONS OF THE CONTRACT FOR CONSTRUCTION, OR AS REQUIRED BY THE ARCHITECTURAL DOCUMENTS AND/OR THE STRUCTURAL ENGINEERS DOCUMENTS ARE PART OF THE CONTRACT.

B. BIDDERS SHALL VISIT AND CAREFULLY EXAMINE THE AREA AFFECTED BY THIS WORK TO FAMILIARIZE THEMSELVES WITH THE EXISTING CONDITIONS AND THE DIFFICULTIES THAT WILL AFFECT THE EXECUTION OF THIS WORK BEFORE SUBMITTING PROPOSALS. SUBMISSION OF A PROPOSAL WILL BE CONSTRUED AS EVIDENCE THAT SUCH AN EXAMINATION HAS BEEN MADE AND LATER CLAIMS WILL NOT BE RECOGNIZED FOR EXTRA LABOR, EQUIPMENT, OR MATERIALS REQUIRED BECAUSE OF DIFFICULTIES ENCOUNTERED WHICH COULD HAVE BEEN FORESEEN HAD SUCH AN EXAMINATION BEEN MADE. ANY DISCREPANCIES SHALL BE BROUGHT TO THE ENGINEERS ATTENTION PRIOR TO BID. IF DISCREPANCIES ARE NOT RESOLVED TO CONTRACTORS SATISFACTION THEY SHALL BE QUALIFIED IN THEIR BID SUBMISSION.

C. THIS CONTRACTOR SHALL REVIEW ALL CONSTRUCTION DOCUMENTS ASSOCIATED WITH THIS PROJECT INCLUDING GENERAL CONSTRUCTION, DEMOLITION, ARCHITECTURAL, MECHANICAL, ELECTRICAL, PLUMBING AND SPRINKLER PLANS AND SPECIFICATIONS. ALL WORK REQUIRED IN THE BID WHICH IS INDICATED OR IMPLIED TO BE PERFORMED BY THIS TRADE IN OTHER SECTIONS OF THE WORK SHALL BE INCLUDED IN THEIR BID. IF A CONFLICT OCCURS IN THE BID SPECIFICATIONS AND/OR ON THE DRAWINGS, THE MORE STRINGENT SITUATION SHALL APPLY.

D. COORDINATE ALL WORK OF THE SECTION WITH EXISTING CONDITIONS AND THE WORK OF OTHER TRADES. THE CONTRACTOR SHALL THOROUGHLY ACQUAINT HIMSELF WITH THE WORK INVOLVED AND SHALL VERIFY AT THE BUILDING ALL MEASUREMENTS NECESSARY FOR THE PROPER INSTALLATION OF THE WORK, OBTAINING THE SAME WHEN NECESSARY FROM THE OTHER CONTRACTORS AND SECTIONS. CONTRACTOR SHALL ALSO BE PREPARED TO PROMPTLY FURNISH TO OTHER CONTRACTORS ANY INFORMATION RELATING TO THE WORK OF THIS SECTION NECESSARY FOR THE PROPER INSTALLATION OF OTHER CONTRACTS AND SHALL COOPERATE TO SECURE THE BEST PROGRESS OF, AND HARMONY BETWEEN, THE WORK OF THE DIFFERENT CONTRACTS AND SECTIONS IN THE INTERESTS OF THE INSTALLATION AS A WHOLE. CONFER WITH OTHER CONTRACTORS AND ENGINEER FOR ADJACENT WORK TO THIS SECTION AND ARRANGE TO HAVE VISIBLE PORTIONS OF WORK FIT AND HARMONIZE IN A MANNER SATISFACTORY TO THE OWNER'S REPRESENTATIVE.

E. THE SPECIFICATIONS ARE ACCOMPANIED BY DRAWINGS INDICATING THE GENERAL LOCATION OF EQUIPMENT AND CONNECTIONS THERETO. UNLESS SPECIFICALLY DIMENSIONED, LOCATIONS OF EQUIPMENT AND ROUTINGS ARE APPROXIMATE. SCALES ON DRAWINGS ARE INDICATED FOR BIDDING PURPOSES ONLY. DRAWINGS SHALL NOT BE SCALED FOR CONSTRUCTION AND MANUFACTURING DETAILS. CERTAIN SYSTEMS ARE DIAGRAMMATIC AND GIVE THE GENERAL ARRANGEMENT ONLY. NO ADDED COMPENSATION WILL BE PERMITTED FOR VARIATIONS DUE TO FIELD CONDITIONS. EXACT LOCATIONS AND ARRANGEMENTS SHALL BE DETERMINED IN THE FIELD ON THE BASIS OF DETAILS INDICATED ON APPROVED SHOP DRAWINGS, AND SUPPLEMENTARY INFORMATION ISSUED BY THE ENGINEER, AND SHALL PROVIDE FOR OPERATING EFFICIENCY, NEATNESS OF APPEARANCE, AND EASE OF MAINTENANCE.

F. GUARANTEE: THE CONTRACTOR SHALL GUARANTEE AND SERVICE THE ENTIRE INSTALLATION FOR A PERIOD OF ONE YEAR FROM THE DATE OF THE FINAL ACCEPTANCE OF THE INSTALLATION. THE CONTRACTOR SHALL, DURING THE PERIOD OF THE GUARANTEE, REPLACE OR REPAIR AT HIS OWN EXPENSE ANY PIECE OF EQUIPMENT AND/OR MATERIAL WHICH IS FOUND TO BE DEFECTIVE. THE REPLACEMENT OR REPAIR SHALL BE PERFORMED THE SAME DAY OF NOTIFICATION IN AN EMERGENCY FASHION WHEN NOTIFIED BY THE OWNER OR AUTHORIZED REPRESENTATIVE. THE CONTRACTOR SHALL ALSO REPAIR ALL DAMAGE TO SURROUNDING WORK CAUSED BY THE FAILURE, REPAIR OR REPLACEMENT OF DEFECTIVE EQUIPMENT. ALL REFRIGERATION COMPRESSORS SHALL HAVE A FACTORY GUARANTEE INCLUDING PARTS AND LABOR FOR FIVE YEARS TOTAL. THE FINAL ACCEPTANCE WILL BE MADE AFTER THE CONTRACTOR HAS ADJUSTED HIS EQUIPMENT, BALANCED THE VARIOUS SYSTEMS, DEMONSTRATED THAT IT FULFILLS THE REQUIREMENTS OF THE DRAWINGS AND SPECIFICATION, AND HAS FURNISHED ALL THE REQUIRED CERTIFICATES OF INSPECTION AND APPROVALS.

G. EQUIPMENT AND MATERIALS: MOST ITEMS OF MECHANICAL AND ELECTRICAL EQUIPMENT AND MATERIAL ARE NOTED ON THE DRAWINGS OR IN THE SPECIFICATIONS WITH A MANUFACTURER'S NAME AND CATALOG NUMBER. THIS DESIGNATION IS USED TO SET THE STANDARD FOR CONSTRUCTION, PERFORMANCE, OPERATION AND APPEARANCE. PRODUCTS OF OTHER MANUFACTURERS WILL BE CONSIDERED AND RULED UPON BY THE ENGINEER. THE SUBMISSION OF A SUBSTITUTION IMPLIES THAT THE ITEM HAS ALL NECESSARY UNDERWRITERS' LABORATORIES, BOARD OF STANDARDS AND APPEALS, NEW YORK CITY MEA, NATIONAL ELECTRICAL CODE, NEW YORK CITY ELECTRICAL CODE AND NEW YORK CITY ELECTRICAL ADVISORY BOARD, ETC. APPROVALS. SHOULD THE ITEM BE FOUND NOT TO HAVE SUCH APPROVAL, IT SHALL BE REPLACED BY THE CONTRACTOR AT NO ADDITIONAL COST TO THE OWNER

H. SUBSTITUTIONS: DEVIATIONS FROM CONTRACT DOCUMENTS AND SUBSTITUTION OF MATERIALS OR EQUIPMENT FOR THOSE SPECIFIED SHALL BE REQUESTED INDIVIDUALLY IN WRITING. FURNISH INFORMATION AS REQUIRED TO DEMONSTRATE THAT THE ARTICLE, MATERIAL, APPARATUS, PRODUCT OR PROCESS TO BE USED IS ADEQUATELY COMPARABLE TO THAT SPECIFIED IN QUALITY, FINISH, DESIGN, EFFICIENCY, DURABILITY AND GENERAL APPEARANCE, AND HAS BEEN ELSEWHERE DEMONSTRATED TO BE SERVICEABLE FOR THE PURPOSES FOR WHICH IT IS INTENDED. IF TESTS OR DEMONSTRATIONS ARE REQUIRED BY THE OWNER'S REPRESENTATIVES, THE COST OF SUCH TESTS OR DEMONSTRATIONS SHALL BE BORNE BY THE CONTRACTOR. DESCRIBE REASON FOR CHANGE, CONNECTIONS TO ADJACENT MATERIALS, ELECTRICAL SERVICES, SERVICE ACCESS REQUIREMENTS, DIFFERENCES IN OPERATING CHARACTERISTICS OR CYCLES AND ALL OTHER POINTS OF DEVIATION. CONTRACTOR TO ASSUME FULL RESPONSIBILITY FOR SAFETY, COORDINATION WITH OTHER TRADES, OPERATION AND PERFORMANCE OF ALTERED SYSTEM.

I. THIS CONTRACTOR IS TO OBTAIN A COPY OF THE BUILDING RULES AND REGULATIONS PRIOR TO BID SUBMISSION. ALL WORK MUST BE INSTALLED IN ACCORDANCE WITH THE BUILDING RULES AND REGULATIONS. DETERMINE REQUIREMENTS AND THE EXTENT OF PREMIUM TIME WORK REQUIRED BY BUILDING, FOR THE PURPOSE OF THE BID ASSUME ANY NOISY WORK (E.G., CHOPPING, CORE DRILLING, WELDING, BRAISING, SOLDERING, ETC.) AND BASE BUILDING SYSTEMS INTERRUPTIONS ARE TO BE PERFORMED OUTSIDE NORMAL BUSINESS HOURS.

J. REMOVAL, TEMPORARY CONNECTIONS AND RELOCATION OF CERTAIN EXISTING WORK WILL BE NECESSARY FOR THE INSTALLATION OF THE NEW SYSTEMS. ALL EXISTING CONDITIONS ARE NOT COMPLETELY DETAILED ON THE DRAWINGS. THE CONTRACTOR SHALL SURVEY THE SITE AND MAKE ALL NECESSARY CHANGES REQUIRED BASED ON EXISTING CONDITIONS FOR PROPER INSTALLATION OF NEW WORK.

K. ALL NECESSARY CUTTING AND PATCHING IN FLOOR SLABS, ROOF SLABS, WALLS, AND CEILINGS FOR THE HVAC WORK SHALL BE PERFORMED BY THIS CONTRACTOR. RESTORE TO MATCH EXISTING CONDITIONS.

L. WHERE PIPE AND/OR DUCTWORK PENETRATE RATED WALLS, THE SPACE BETWEEN THE INSULATION AND THE WALL SHALL BE CAULKED WITH NON-COMBUSTIBLE MATERIAL IN AN APPROVED MANNER. ALL PIPING AND/OR DUCTWORK TO BE INSTALLED ABOVE HUNG CEILING UNLESS OTHERWISE NOTED ON DRAWINGS. THE CONTRACTOR SHALL COORDINATE WITH ARCHITECTURAL DRAWINGS FOR ALL CEILING ELEVATIONS.

M. ACCESS DOORS IN FINISHED CONSTRUCTION: THE CONTRACTOR SHALL PREPARE A LIST OF ALL ACCESS DOORS (MINIMUM 18"X18") REQUIRED FOR OPERATION AND MAINTENANCE OF ALL CONCEALED EQUIPMENT AND OTHER DEVICES, WHICH SHALL BE SUPPLIED TO THE GENERAL CONTRACTOR FOR INSTALLATION. THE COST TO FURNISH AND INSTALL ACCESS DOORS SHALL BE INCLUDED IN THIS CONTRACTORS BID. THIS CONTRACTOR IN ADVANCE OF CEILING INSTALLATIONS SHALL SUITABLY FIELD TAG AND IDENTIFY ALL CONCEALED EQUIPMENT, VALVES, DAMPERS, ETC., WHICH REQUIRE ACCESS DOOR PROVISIONS.

N. NEW DUCTWORK SHALL ARRIVE ON THE CONSTRUCTION SITE SEALED AND REMAIN PROTECTED FROM DEBRIS THROUGHOUT CONSTRUCTION PRIOR TO FINAL INSTALLATION. AIR DISTRIBUTION ACCESSORIES AND INTERNAL COMPONENTS OF ALL HVAC EQUIPMENT SHALL BE SEALED AND PROTECTED FROM DEBRIS WHILE ON THE CONSTRUCTION SITE PRIOR TO FINAL CONNECTION AND START-UP.

O. ALL VOLATILE ORGANIC COMPOUND (VOC) LIMITS OF ADHESIVES, SEALANTS AND SEALANT PRIMERS MUST COMPLY WITH SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD) RULE #1168, AMENDMENT DATE OF JANUARY 7, 2005.

1.02 SCOPE OF WORK

A. THE CONTRACTOR SHALL FURNISH AND INSTALL AN HVAC SYSTEM COMPLETE WITH ALL EQUIPMENT, DUCTWORK, PIPING, INSULATION, CONTROLS, ACCESSORIES AND ASSOCIATED WORK IN ACCORDANCE WITH THE NEW YORK CITY BUILDING CODE, ALL NATIONAL, STATE AND LOCAL AUTHORITIES HAVING JURISDICTION, BUILDING MANAGEMENT, DESIGN DRAWINGS AND THIS SPECIFICATION.

B. THE WORK SHALL INCLUDE ALL LABOR, MATERIALS, EQUIPMENT, HOISTING AND RIGGING, BREAKDOWN AND SETUP OF EQUIPMENT FOR INSTALLATION, SCAFFOLDING, AND SERVICES TO COMPLETE THE SYSTEM AND PROVIDE THE OWNER WITH A FULLY OPERATIONAL SYSTEM. ANY EQUIPMENT, PARTS, MATERIALS, ACCESSORIES, OR LABOR THAT IS NECESSARY FOR PROPER PERFORMANCE OF THE MECHANICAL WORK ALTHOUGH NOT SPECIFICALLY MENTIONED HEREIN OR SHOWN ON THE DRAWINGS, SHALL BE FURNISHED AND INSTALLED WITHOUT ADDITIONAL COSTS. WHEN INSTALLATION OF A PART OF ANY SYSTEM (PLUMBING, HEATING, AIR CONDITIONING, ELECTRICAL OR OTHERWISE) REQUIRES A SHUTDOWN OF ANY OPERATING SYSTEM, CONNECT THE PARTIAL SYSTEM ONLY AFTER NOTIFICATION TO AND WITH APPROVAL OF THE OWNER. COORDINATE ACTIVITIES CLOSELY WITH THOSE OF SUBCONTRACTOR'S SO THE OPERATION IS RESTRICTED TO AS SHORT AN INTERVAL AS POSSIBLE AND "OUT OF SERVICE" TIME OF THESE FACILITIES IS KEPT TO A MINIMUM. ANY SHUTDOWN OF THE ELECTRICAL SYSTEM WILL BE DONE OUT OF HOURS AS APPROVED BY OWNER.

C. IT IS IMPERATIVE THAT EXISTING SYSTEMS BE MAINTAINED IN CONTINUOUS OPERATION DURING THE COURSE OF CONSTRUCTION; IF SHUTDOWNS ARE REQUIRED TO PERMIT THE DISCONNECTION AND REMOVAL OR RECONNECTION OF EXISTING WORK, OR FINAL CONNECTION TO BE MADE TO AN EXISTING SYSTEM, THEY SHALL OCCUR ONLY DURING OFF-HOURS AND ONLY AFTER PROPER PERMISSION HAS BEEN OBTAINED FROM BUILDING MANAGEMENT.

D. THE BUILDING MANAGEMENT REQUIRES NOT LESS THAN SEVEN DAYS NOTICE FOR SHUTDOWN OF ANY BUILDING SYSTEM.

E. MAKE AN ACCURATE TAKE-OFF ALL EXISTING EQUIPMENT, DUCTWORK, PIPING, CONDUIT, PANELBOARDS, WIRING DEVICES, AND OTHER ACCESSORIES BEING REMOVED DURING DEMOLITION AND INCLUDE THE COST FOR DISCONNECTING AND REMOVAL OF STATED EQUIPMENT, ETC. INTO THE BASE BID. REMOVALS SHALL BE AS SPECIFIED AND/OR AS INDICATED ON THE DRAWINGS. IN CERTAIN CASES, EQUIPMENT OR MATERIALS DESIGNATED FOR REMOVAL SHALL REMAIN THE PROPERTY OF THE OWNER AND SHALL BE TURNED OVER AT LOCATIONS IN THE BUILDING AS DIRECTED BY THE OWNER.

F. PLAN INSTALLATION OF NEW WORK AND CONNECTIONS TO EXISTING WORK TO INSURE MINIMUM INTERFERENCE WITH REGULAR OPERATION OF EXISTING FACILITIES. ALL SYSTEM SHUTDOWNS AFFECTING OTHER AREAS SHALL BE COORDINATED WITH BUILDING MANAGEMENT.

G. THIS TENANT/OWNER SHALL PROCURE THE SERVICES OF A THIRD PARTY INSPECTION COMPANY TO PERFORM ALL SPECIAL INSPECTIONS IN ACCORDANCE WITH THE NEW YORK CITY BUILDING CODE. SECURE ALL REQUIRED PERMITS AND APPROVALS AND TRANSMIT SAME TO THE OWNER. CONTRACTOR SHALL BE RESPONSIBLE FOR ALL FEES.

1.03 SHOP DRAWINGS, EQUIPMENT SUBMISSION, MAINTENANCE MANUALS

A. SUBMIT ONE (1) REPRODUCIBLE AND ONE (1) PRINT OF THE SHEET METAL AND PIPING SHOP DRAWINGS, 3/8" SCALE, CERTIFIED BY ALL TRADES THAT COORDINATION HAS BEEN ESTABLISHED.

B. SUBMIT THREE (3) COPIES OF ALL SHEET METAL AND PIPING SHOP STANDARDS LEAKAGE TEST CERTIFICATION, AIR AND WATER BALANCING REPORTS, AND CERTIFIED EQUIPMENT CUTS WITH CONSTRUCTION WIRING DIAGRAMS, AND AUTOMATIC TEMPERATURE CONTROL SHOP DRAWINGS INCLUDING CONTROL AND POWER WIRING DIAGRAMS, SEQUENCE OF OPERATIONS AND ALL CUTS OF EQUIPMENT AND DEVICES.

C. SUBMIT FOUR (4) BOOK BOUND OPERATING AND SERVICE MANUALS WHICH SHALL INCLUDE COPIES OF ALL AS-BUILT SHOP DRAWINGS FOLDED AND PLACED INTO BINDER POCKETS, AS-BUILT DRAWINGS IN ELECTRONIC FORMAT, COPIES OF REVIEWED EQUIPMENT CUTS FOR INSTALLED EQUIPMENT, COPIES OF EQUIPMENT START UP CHECKLISTS, AIR AND WATER BALANCING REPORTS, LEAK TESTS, HYDROSTATIC TESTS, WATER TREATMENT AND CHEMICAL CLEANING CERTIFICATION. CONTRACTOR SHALL INSTRUCT OWNERS PERSONNEL ON THE OPERATION OF ALL HVAC SYSTEMS.

D. AS WORK PROGRESSES AND FOR DURATION OF CONTRACTOR, MAINTAIN COMPLETE AND SEPARATE SET OF PRINTS OF CONTRACT DRAWINGS AT THE JOB SITE. RECORD WORK COMPLETED AND ALL CHANGES FROM ORIGINAL CONTRACT DRAWINGS CLEARLY AND ACCURATELY INCLUDING WORK INSTALLED AS A MODIFICATION OR ADDITION TO THE ORIGINAL DESIGN. RECORD VALVE TAGS AS THEY ARE INSTALLED. FINAL SUBMISSION OF REPRODUCIBLE AS-BUILT DRAWINGS ARE TO BE SIGNED AND CERTIFIED BY INSTALLING CONTRACTOR THAT THIS IS THE AS-BUILT CONDITION OF THE WORK. AS-BUILT SHOP DRAWINGS SHALL BE SUBMITTED IN DRAWING AND ELECTRONIC FORMAT (AUTOCAD 2007 MINIMUM).

PART 2- PRODUCT/APPLICATION

2.01 DUCTWORK

A. PROVIDE ALL SUPPLY, RETURN, EXHAUST, AND OUTSIDE AIR SHEET METAL DUCTWORK, FITTINGS, DAMPERS, TURNING VANES, ACCESS DOORS, PLENUMS, FLEXIBLE CONNECTIONS, AND SUPPORTS AND PERFORM LEAK TEST PER LATEST SMACNA STANDARDS AND NFPA90A AS MODIFIED BY N.Y.C. BUILDING CODE. ALL DUCTWORK JOINTS SHALL BE SEALED AIR TIGHT WITH APPROVED DUCT SEALANT, SIMILAR TO 3M-900.

B. ALL LOW PRESSURE DUCTS EXPOSED IN OCCUPIED AREAS, OTHER THAN MECHANICAL AND FAN ROOMS FABRICATED WITH HEMMED "S" SLIPS. REINFORCE JOINTS OF DUCTS OVER 30" WIDE WITH FLAT BARS OR FLAT BARS AND 3/8" RODS FOR DUCTS OVER 54" WIDE. TOP JOINT WITH BAR SKIP UNDER 31" WIDTH AND REINFORCED BAR SKIP FOR 31" AND LARGER IN WIDTH.

C. ROUND DUCTS SPIRAL LOCK. G.I. COMPANY, SHEET METAL PRODUCTS, UNITED SHEET METAL, PACIFIC AIR PRODUCTS, OR AS APPROVED. ROUND DUCTS OVER 60" WITH BUTT WELDED, LONGITUDINAL SEAMS, AND FLANGE JOINTS.

D. FITTINGS IN ROUND DUCTS SHALL BE NO LIGHTER THAN 20 GAUGE, AND WELDED. G. 1. COMPANY, SHEET METAL PRODUCTS, UNITS SHEET METAL, PACIFIC AIR PRODUCTS, OR AS APPROVED. BRANCH TEE TAKE-OFFS MADE WITH "CON-T" TYPE CONICAL TEE FITTINGS. WHERE MAIN DUCT REDUCES IN SIZE AFTER TAKE-OFF, USE "CON-T" OR TURNS, AND 3-PIECE FOR 45 DEGREE TURNS.

E. CONTRACTOR SHALL ADHERE TO THE FULL INSIDE CROSS SECTIONAL DUCTWORK AREAS SHOWN ON THE DRAWINGS AND PROVIDE ALL TRANSITIONS AND OFFSETS AS REQUIRED TO MEET FIELD CONDITIONS, ACCOMMODATE EQUIPMENT MAINTENANCE REQUIREMENTS AND COORDINATE WITH ALL TRADES. ALL FIELD CONDITIONS WHICH REQUIRE MODIFIED TRANSITIONS WILL NOT BE APPROVED WITHOUT PRIOR ENGINEER APPROVAL THROUGH SHOP DRAWING OR RFI.

F. FOR DUCTS WITH ACOUSTICAL LINING THE SIZES SHOWN ON THE PLAN SHALL BE THE CLEAR INSIDE DIMENSIONS.

G. ALL OPEN-ENDED RETURN, TRANSFER OR EXHAUST DUCTS SHALL BE PROVIDED WITH WIRE MESH SCREENS.

H. NEW AND EXISTING DUCTWORK TO BE REUSED SHALL HAVE PRESSURE CLASSIFICATION, SEALING REQUIREMENTS AND LEAKAGE TESTING AS LISTED BELOW UNLESS OTHERWISE SPECIFIED OR SHOWN ON THE DRAWINGS:

1. 4" CLASS: ALL SUPPLY DUCTWORK FROM DISCHARGE OF AIR UNITS TO INLETS OR TERMINAL BOXES. SEAL CLASS A, LEAKAGE CLASS 6 (RECTANGULAR) OR CLASS 3 (ROUND). PROVIDE TDF FLANGE CONNECTIONS FOR ALL SYSTEM 4" PRESSURE CLASS AND ABOVE.
2. 2" CLASS: ALL OTHER LOW PRESSURE DUCTWORK. SEAL CLASS B, LEAKAGE CLASS 24 (RECTANGULAR) OR CLASS 12 (ROUND).
3. LEAKAGE TESTING:
   - A) FOR 4" CLASS DUCTWORK, ALL DUCTWORK SHALL BE TESTED TO VERIFY THAT ACTUAL AIR LEAKAGE IS BELOW ALLOWABLE AMOUNTS AS DEFINED BY ASHRAE 90.1 OR THE ENERGY CODE AS INDICATED ON THE PROJECT'S COMCHECK COMPLIANCE CERTIFICATE, WHICHEVER IS MORE STRINGENT. TESTING SHALL INCLUDE ALL NEW DUCTWORK UPSTREAM OF FAN POWERED TERMINAL UNITS INCLUDING, BUT NOT LIMITED TO MAINS, BRANCHES, AND THE PRIMARY AIR INLET BRANCHES TO EACH FP-VAV. WORK SHALL BE WITNESSED BY THE BUILDING'S CHIEF ENGINEER OR HIS REPRESENTATIVE AND A COPY OF THE CERTIFIED TESTING REPORT SHALL BE SUBMITTED TO THE BUILDING MANAGEMENT.
   - B) ALL NEW LOW PRESSURE DUCTWORK (2" CLASS) SHALL BE TESTED ON AN AS-NEEDED BASIS AT THE ENGINEERS DISCRETION OR IF BALANCING AIR QUANTITIES CAN NOT BE MET. IF SPECIMEN FAILS TO MEET ALLOTTED LEAKAGE LEVEL, THE CONTRACTOR SHALL MODIFY TO BRING IT INTO COMPLIANCE AND SHALL RETEST IT UNTIL ACCEPTABLE LEAKAGE IS DEMONSTRATED. TESTS AND NECESSARY REPAIRS SHALL BE COMPLETED PRIOR TO CONCEALMENT OF DUCTS.

F. MATERIALS:

1. SHEETMETAL: HOT-DIPPED GALVANIZED SHEETMETAL WITH G60 COMMERCIAL COATING ACCORDING TO ASTM A653 & A924 FOR ALL DUCTWORK UNLESS OTHERWISE SPECIFIED.
2. ALUMINUM: ALLOY 3003-H14, OF THICKNESS REQUIRED BY THE SMACNA DUCT CONSTRUCTION STANDARDS. PROVIDE FOR ALL DUCTWORK EXPOSED TO WEATHER AND MOISTURE INCLUDING OUTSIDE AIR DUCTS WITHIN 10 FEET OF LOUVERS AND TOILET ROOMS EQUIPPED WITH BATHS OR SHOWERS.
3. FLEXIBLE CONNECTIONS AT FANS SHALL BE NEOPRENE COATED, FLAME RETARDANT GLASS FABRIC (COMPLYING WITH NFPA 90), 30 OZ./SQ, YD. WITH SEWED AND CEMENTED SEAMS.

G. PROVIDE MANUAL BALANCING DAMPERS AS REQUIRED TO PROPERLY BALANCE THE AIR DISTRIBUTION SYSTEM AS SHOWN ON DRAWINGS AND AS LISTED BELOW:

1. ALL SUPPLY AIR MAIN BRANCHES FROM TRUNK, EACH SPLIT, AND ALL SUB-BRANCHES FROM MAINS SHALL HAVE BALANCING DAMPERS.
2. EXHAUST AND RETURN MAIN BRANCHES FROM TRUNK, EACH SPLIT AND ALL SUB-BRANCHES FROM MAINS SHALL HAVE BALANCING DAMPERS.
3. IF DAMPER IS NOT ACCESSIBLE, OR IS LOCATED ABOVE A PLASTER OR DRYWALL CEILING, PROVIDE A REMOTE DAMPER ACTUATOR AND DAMPER AS MANUFACTURED BY YOUNG REGULATOR MODEL 896-C WITH NO. 1200A RIGHT ANGLE WORM GEAR AND DAMPER MODEL 820 OR APPROVED EQUAL.

H. FIRE DAMPERS AND FIRE/SMOKE DAMPERS:

1. PROVIDE ALL FIRE DAMPERS, SMOKE DAMPERS, COMBINATION FIRE/SMOKE DAMPERS, SMOKE DETECTORS, AND ASSOCIATED CONTROLS AND ALARMS AS REQUIRED BY CODE.
2. DAMPERS SHALL BE DYNAMIC TYPE, U.L. LISTED AND LABELED, AND IN CONFORMANCE WITH NFPA.
3. FIRE DAMPER SHALL BE FUSIBLE LINK TYPE (165 DEGREE F.), TYPE B SHUTTER OUT OF THE AIR STREAM AS MANUFACTURED BY POTTORFF MODEL VFD-10 (1-1/2 HR RATED) OR MODEL VFD-30 (3HR RATED) AS REQUIRED OR APPROVED EQUAL.
4. COMBINATION FIRE/SMOKE DAMPERS SHALL BE CLASS II, REMOTE RESETTABLE, MULTI-BLADE TYPE WITH FIRESTAT OR HEAT SENSOR DEVICE AND 120-VOLT ACTUATOR MOUNTED OUT OF THE AIR STREAM, WITH DAMPER OPERATOR AND BLADE POSITION INDICATOR SWITCHES. PROVIDE MOTOR MOUNT BRACKET STRENGTHENER FOR DAMPERS OVER 10" IN HEIGHT. PROVIDE A 10 GAUGE WELDED VERTICAL STIFFENER AT EACH CORNER TO PREVENT DAMPER MISALIGNMENT.
5. COMBINATION FIRE/SMOKE DAMPERS SHALL BE MANUFACTURED BY POTTORFF MODEL FSD-142 (1-1/2 HR RATED) OR MODEL FSD-342 (3HR RATED) AS REQUIRED OR APPROVED EQUAL.
6. THE HVAC CONTRACTOR SHALL PROVIDE ALL DEVICES, RELAYS, END SWITCHES, E/P SWITCHES, CONTROL COMPONENTS, AIR PIPING, POWER WIRING, CONTROL WIRING AND INTERLOCK WIRING AS REQUIRED TO ACCOMPLISH THE SEQUENCE OF OPERATION FOR THESE DAMPERS.
7. DUCTWORK SHALL BE TRANSITIONED LARGER AT ALL FIRE AND FIRE/SMOKE DAMPERS SUCH THAT THE NET FREE AREA OF THE DUCTWORK IS NOT COMPROMISED.
8. CONTRACTOR TO PROVIDE A 12"X12" ACCESS DOOR IN THE DUCTWORK FOR ACCESS TO INTERNAL COMPONENTS OF THE DAMPER.

I. PROTECTION PANS AND DRIP PANS:

1. PROVIDE PROTECTION PANS UNDER NEW CEILING MOUNTED AIR CONDITIONING UNITS, PIPES PASSING THROUGH SWITCHGEAR ROOMS OR OVER ELECTRIC EQUIPMENT. THE PANS SHALL BE CONSTRUCTED OF GALVANIZED STEEL, SUITABLE REINFORCEMENT TO PREVENT SAGGING.
2. EDGES OF THE PANS SHALL TURN UP 2" ON ALL SIDES WITH CORNERS SEALED TO MAKE PAN WATERTIGHT.
3. PAN SHALL BE SUPPORTED BY PIPE HANGERS AND SHALL DRAIN CLEAR OF ELECTRICAL EQUIPMENT.
4. PROVIDE A 3/4" DRAIN PIPE FOR EACH PAN TERMINATING ABOVE NEAREST CONVENIENT SINK OF FLOOR DRAIN. SLOPE AND DRAIN ALL DUCTS EXPOSED TO MOISTURE, CONSTRUCT OF ALUMINUM AND DO NOT INTERNALLY LINE.

J. SLOPE AND DRAIN ALL DUCTS EXPOSED TO MOISTURE, CONSTRUCT OF ALUMINUM AND DO NOT INTERNALLY LINE

K. AUTOMATIC CONTROL DAMPERS: PROVIDE DAMPERS WITH PARALLEL BLADES FOR 2-POSITION OR MIXING CONTROL, OR OPPOSED BLADES FOR MODULATING CONTROL OF CONSTANT OR VARIABLE VOLUME SYSTEM. AUTOMATIC DAMPERS ARE TO BE VERY LOW LEAKING TYPE WITH A MAXIMUM LEAKAGE RATE OF 6 CFM PER SQUARE FOOT AT 4" W.G. DAMPER MATERIAL SHALL BE THE SAME AS DUCT. PROVIDE WEATHERPROOF COMPONENTS FOR DAMPERS IN A MOISTURE ENVIRONMENT.

2.02 GRILLES, REGISTERS AND DIFFUSERS

A. PROVIDE ALL AIR OUTLETS AND RETURNS OF THE TYPE AND SIZES, AS SELECTED AND INDICATED ON DRAWING. ALL DUCTED RETURN AND EXHAUST OUTLETS SHALL HAVE OPPOSED BLADE DAMPERS (ADJUSTABLE THROUGH THE FACE). PROVIDE FRAMES AND MOUNTING TYPES AS REQUIRED TO MATCH SURROUNDING CEILING CONSTRUCTION. FINISHES TO BE SELECTED BY THE ARCHITECT.

B. ALL CEILING TYPE AIR DIFFUSERS SHALL BE PROVIDED WITH EQUALIZING DEFLECTOR.

C. A SCHEDULE OF DIFFUSERS, GRILLES AND REGISTERS WITH MANUFACTURERS MODELS, SIZES, ACCESSORIES, FINISHES, ETC., SHALL BE SUBMITTED FOR APPROVAL PRIOR TO RELEASE FOR FABRICATION AND DELIVERY

D. DIFFUSERS SHOWN ON DIFFUSER SCHEDULE SHALL BE CHANGED TO MATCH EXISTING DIFFUSER TYPE WHERE EXISTING DIFFUSERS ARE REMAINING.

E. ALL LINEAR DIFFUSERS SHALL BE PROVIDED WITH PATTERN CONTROL VANES. ALL ADJUSTABLE PATTERN DEFLECTORS SHALL BE FIELD ADJUSTED TO OPTIMIZE AIR DISTRIBUTION PREVENTING DRAFT CONDITIONS. CONTRACTOR SHALL PLAN FOR A SECOND COMFORT FIELD ADJUSTMENT PER OWNER/ENGINEER DISCRETION.

F. ALL UNUSED PORTIONS OF SUPPLY LINEAR DIFFUSERS SHALL BE USED AS RETURN GRILLE AND SHOULD BE PROVIDED WITH LIGHT SHIELDS.

G. ALL LINEAR DIFFUSERS AND DIFFUSERS IN INACCESSIBLE CEILINGS SHALL BE PROVIDED WITH A REMOTE OPERATED OPPOSED BLADE DAMPER AND A 3-FOOT (MINIMUM) FLEXIBLE ADJUSTMENT CABLE WITH 1/8-INCH KEY OPERATOR.

2.03 PIPING

A. PROVIDE PIPING WHICH IS SCHEMATICALLY INDICATED AND SIZED ON DRAWINGS. PIPING TO BE INSTALLED TO MEET SPECIFIED HEADROOM OR FIELD CONDITIONS AND SHALL CONFORM TO LATEST ASME CODES FOR PRESSURE PIPING. PIPE MATERIALS AND FITTING MATERIALS SHALL BE AS PER THE PIPE AND FITTING SCHEDULES SHOWN ON DRAWINGS.

B. PIPING, FITTINGS, AND ALL PIPE APPURTENANCES SHALL BE SUITABLE FOR THE PRESSURE AND TEMPERATURE OF SERVICE.

C. PROVIDE DIELECTRIC FITTINGS TO CONNECT DIFFERENT PIPING MATERIALS.

D. PROVIDE AIR VENTS AT EACH HIGH POINT AND DRAIN VALVES WITH HOSE BIB AT EACH LOW POINT.

E. PIPING SHALL BE INSTALLED WITH PROPER ANCHORS AND EXPANSION/CONTRACTION DEVICES SUCH AS LOOPS OR APPROVED EXPANSION JOINTS TO PREVENT UNDUE STRAINS ON PIPING OR APPARATUS CONNECTED TO THE PIPING, AS REQUIRED.

F. SUPPORT PIPING WITH HANGERS EQUIPPED WITH INSULATION SADDLES FROM APPROVED CONCRETE INSERTS, EXPANSION SHIELDS, BEAM CLAMPS, AND/OR SUPPLEMENTARY STEEL ANGLES, PLATES, AND CHANNELS. CONTRACTOR SHALL SUBMIT METHOD OF PIPING SUPPORT SIGNED AND SEALED BY A LICENSED PROFESSIONAL ENGINEER FOR REVIEW.

G. UNIONS WITH REMOVABLE SECTIONS OF PIPING SHALL BE INSTALLED AT ALL EQUIPMENT TO PERMIT EASE OF DISCONNECTION FOR EQUIPMENT SERVICE/REMOVALS WITHOUT DISMANTLING OF MAJOR PORTIONS OF CONNECTED PIPING.

H. PROVIDE TEES IN PIPING SYSTEM FOR TESTING AND BALANCING, AND INSTALLATIONS OF FLOW OR FLOAT SWITCHES, GAUGED, THERMOMETERS AND OTHER BALANCING AND CONTROL DEVICES, COORDINATE WITH THE CONTROL CONTRACTOR AND BALANCER.

I. PROVIDE AUTOMATIC PRESSURE RELIEF VALVES AND VACUUM BREAKERS TO PREVENT AGAINST PIPE RUPTURE OR SYPHONING ACTIONS. EXTEND DRAINS FROM RELIEF VALVES TO SPILL OVER FLOOR DRAINS.

J. ALL PIPE SLEEVES SHALL BE SCHEDULE 40 GALVANIZED STEEL. ANNULUS BETWEEN PIPE OR PIPE INSULATION AND SLEEVE SHALL BE CAULKED WITH A NON-COMBUSTIBLE MATERIAL TO WITHIN 1/4" OF WALL FACES AND FILLED WITH CAULKING COMPOUND FOR INTERIOR SLEEVES. EXTERIOR SLEEVES OR WATERPROOF SLEEVES SHALL UTILIZE LINK SEAL (LS) TYPE TO FILL THE ANNULUS. PROVIDE ESCUTCHEONS ON ALL EXPOSED PIPING THROUGH WALLS OR FLOORS HELD IN PLACE WITH SCREWS.

K. PROVIDE SECURELY FASTENED LABELING OF ALL PIPING (BOTH EXPOSED AND CONCEALED) IN ACCORDANCE WITH ANSI STANDARDS AND COLOR CODED AS PER BUILDING MANAGEMENT STANDARDS. LABELING SHOULD BE PROVIDED 20 FEET ON CENTERS AND/OR AT LEAST ONCE IN EACH ENCLOSED SPACE OR ROOM WHERE THE WALLS EXTEND ABOVE THE CEILING.

L. PROVIDE VALVE TAGS AND CHARTS:

1. EACH VALVE SHALL HAVE A 2 INCH DIAMETER BRASS TAG WITH 1 INCH HIGH NUMERAL STAMPED THEREON, SECURED TO THE VALVE BY MEANS OF BRASS S HOOK OR BRASS CHAIN. EACH SYSTEM TO HAVE A LETTER DESIGNATION INDICATING SERVICE.
2. THE CONTRACTOR SHALL FURNISH AN APPROVED NEATLY DRAWN VALVE CHART, PROPERLY FRAMED, SHOWING THE USE AND LOCATION OF EACH VALVE THAT IS TAGGED.

M. VALVES AND STRAINERS:

1. VALVES, STRAINERS, STEAM TRAPS, ETC., SHALL NOT CONTAIN ASBESTOS AND HAVE THE NAME OF THE MANUFACTURER AND GUARANTEED WORKING PRESSURE CAST OR STAMPED ON BODIES. VALVES OF SIMILAR TYPE SHALL BE BY A SINGLE MANUFACTURER.
2. VALVES USED FOR THROTTLING OR CONTROLLING FLOW SHALL BE BALL (3" OR SMALLER) OR PLUG TYPE VALVES (ALL SIZES). VALVES FOR ISOLATION SHALL BE BALL FOR LIQUID SYSTEMS AND GATE FOR STEAM SYSTEMS UNLESS OTHERWISE SPECIFIED. BUTTERFLY VALVE SHALL BE LUG TYPE AND MAY BE SUBSTITUTED FOR BALL VALVES FOR SIZES 4" AND LARGER. BUTTERFLY VALVES SHALL NOT BE USED FOR MODULATING SERVICE OR STEAM SERVICE, USE ONLY FOR 2 POSITION ISOLATION ON WATER SYSTEMS. REFER TO AUTOMATIC TEMPERATURE CONTROL SECTION FOR CONTROL VALVES.
3. VALVES SHALL HAVE WORKING PRESSURE AND TEMPERATURE RATINGS SAME AS PIPE FITTINGS SPECIFIED FOR THE SERVICE. REGARDLESS OF SERVICE, VALVES SHALL NOT BE DESIGNED FOR LESS THAN 300 PSI COLD WORKING PRESSURE.
4. LUBRICATED, TAPERED PLUG VALVES WITH LOCKING FLOW PLATE SHALL BE PROVIDED IN THE DISCHARGE PIPING FROM WATER CIRCULATING PUMPS, IN THE LEAVING WATER PIPING BRANCHES FROM ALL COILS, HEAT EXCHANGER TYPES OF EQUIPMENT, AND ALL RETURN WATER RISERS OF SUB-MAINS THAT CONNECT TO HYDRONIC MAINS FOR BOTH BALANCING AND ISOLATION PURPOSES.
5. CHECK VALVES SIZED 2-1/2" AND LARGER SHALL BE IRON BODY, FLANGED ENDS, BRONZE MOUNTED, SWING PATTERN, WITH REMOVABLE CAP, RE-GRINDABLE DISC AND SEAT RING. CHECK VALVES SIZED 2" AND SMALLER SHALL BE BRONZE BODY, SCREWED ENDS, SWING PATTERN. PROVIDE SPRING LOADED, SILENT ACTION, NON-SLAM TYPE CHECK VALVE WITH REMOVABLE CAP, RE-GRINDING DISC AND SEAT RING IN ALL VERTICAL INSTALLATIONS AND DISCHARGE PIPING FROM PUMPS AS MANUFACTURED BY SMOLENKSY, MEULLER, WILLIAMS-HAGER OR MILLER.
6. BALL VALVES SHALL BE PROVIDED WITH STAINLESS STEEL BALL, STEM AND SEAT RING, TFE BUSHING AND SEAT RING GASKET. BALL VALVES INSTALLED IN COPPER SYSTEMS SHALL HAVE BRONZE BODIES. BALL VALVES SHALL BE RATED FOR A MINIMUM OF 275 PSI @ 100 DEGREE F. BALL VALVES USED FOR THROTTLING (3" AND SMALLER) SHALL BE PROVIDED WITH A LOCKING BALANCING STOP.
7. STRAINERS OF SARCO OR MEULLER MANUFACTURER SHALL BE PROVIDED IN THE INLET PIPING TO EACH STEAM TRAP, MAKE UP CONNECTION, PUMP, AND AUTOMATIC CONTROL VALVE OF STEAM AND HYDRONIC SYSTEM. STRAINER SHALL BE Y-PATTERN UNLESS OTHERWISE SPECIFIED ON DRAWINGS. STRAINERS SHALL BE OF DESIGN TO ALLOW BLOW-DOWN OF ACCUMULATED DEBRIS AND TO FACILITATE REMOVAL AND REPLACEMENT OF THE STRAINER SCREEN WITHOUT DISCONNECTION FROM THE MAIN PIPING. STRAINERS INSTALLED IN COPPER SYSTEMS SHALL HAVE BRONZE BODIES. STRAINER BASKET SHALL BE NICKEL, COPPER, BRASS OR STAINLESS STEEL OF AMPLE STRENGTH TO PREVENT COLLAPSING UNDER SHOCK LOADING. PERFORATIONS SHALL BE AS FOLLOWS: STEAM=1/32", WATER UP TO 3" SIZE-1/16", WATER 4" AND OVER -1/8". FOR STRAINERS 2-1/2" AND LARGER, PROVIDE A VALVE DIRT BLOW-OUT PIPING CONNECTION TERMINATED WITH A PIPE NIPPLE AND CAP. STRAINERS 2" AND SMALLER SHALL HAVE 6" LONG BLOW-OFF NIPPLE WITH CAPPED END.

N. THERMOMETERS AND PRESSURE GAUGES:

1. PROVIDE PIPE THERMOMETERS WITH SEPARABLE SOCKETS IN THE ENTERING AND LEAVING WATER PIPING CONNECTIONS TO COOLING TOWERS, CHILLERS, HEAT EXCHANGES, HEATING, COOLING AND CONDENSER COILS. THERMOMETERS SHALL BE WEISS, WEKSLER, THERICE OR OTHER APPROVED MANUFACTURER AND SHALL BE MINIMUM OF 4-1/2" DIAL TYPE, ALUMINUM FLANGELESS CASE FURNISHED WITH MICROMETER ADJUSTABLE POINTER. THERMOMETER SHALL HAVE A 1% ACCURACY AND MIDPOINT AS SYSTEM OPERATING TEMPERATURE.
2. PROVIDE LIQUID FILLED PRESSURE GAUGES ON INLET AND OUTLET WATER PIPING CONNECTIONS TO ALL PUMPS AND OTHER WATER HEAT EXCHANGE APPARATUS INCLUDING WATER COILS, HEAT EXCHANGERS, CHILLERS. EACH PRESSURE GAUGE INSTALLATION SHALL INCLUDE A 1/4" BALL VALVE FOR ITS CONNECTION TO PIPING. PRESSURE GAUGES SHALL BE WEISS, WEKSLER, THERICE OR OTHER APPROVED MANUFACTURER AND SHALL BE MINIMUM OF 4-1/2" DIAL TYPE, CAST ALUMINUM CASE, STEEL MOVEMENT, MICROMETER ADJUSTABLE POINTER, 1% ACCURACY AND MIDPOINT AT SYSTEM OPERATING PRESSURE.

O. PIPE TESTING:

1. NO TESTING SHALL BE CONDUCTED UNTIL PIPE CLEANING AND PRETREATMENT HAS BEEN COMPLETED AND RECORDED.
2. ALL TESTING SHALL BE COORDINATED BY THE CONTRACTOR AND SHALL BE WITNESSED BY A BUILDING OWNERS REPRESENTATIVE. ALL SYSTEMS WHICH FAIL THE PRESSURE TESTS SHALL BE FIXED AND RETESTED AT NO EXPENSE TO THE OWNER.
3. ISOLATE ALL EQUIPMENT WHICH IS TO BE EXCLUDED FROM THE PRESSURE TEST AND PROVIDE ALL TEMPORARY PIPING CONNECTIONS, FITTINGS, VALVES, EQUIPMENT, LABOR, ETC., TO PRESSURE TEST ALL SYSTEMS.
4. ALL TENANT WATER SYSTEM SHALL BE ISOLATED FROM THE BASE BUILDING SYSTEM.
5. CONDENSER AND HOT WATER SYSTEMS WILL BE HYDROSTATICALLY TESTED WITH WATER AT 1-1/2 TIMES THE WORKING PRESSURE, FOR A MINIMUM PERIOD OF FOUR HOURS, WITH NO LEAKS.

2.04 INSULATION REQUIREMENTS

A. INSULATION SHALL BE APPLIED TO PIPING AND DUCTWORK OF MATERIALS AS SPECIFIED HEREIN AND FOR APPLICABLE SYSTEMS OF THIS PROJECT. INSULATION SHALL HAVE A FLAME SPREAD RATING NOT EXCEEDING 25 AND A SMOKE DEVELOPED INDEX OF 50 OR LESS AND SHALL MEET THE REQUIREMENTS OF ASTM, NFPA.

B. INSULATION SHALL BE CONTINUOUS THROUGH WALL AND SLAB SLEEVE OPENINGS EXCEPT FOR RATED WALLS OR SLABS WHERE AN APPROVED FIRESTOP IS REQUIRED AS PER NFPA.

C. INSULATION OF COLD SURFACES WHERE VAPOR BARRIER JACKETS ARE SPECIFIED SHALL BE APPLIED WITH AN UNBROKEN VAPOR SEAL. HANGERS AND SUPPORTS THAT ARE SECURED TO COLD SURFACES SHALL BE ADEQUATELY INSULATED TO PREVENT CONDENSATION.

D. WHERE INSULATION IS SPECIFIED FOR PIPING, INSULATE SIMILARLY ALL CONNECTIONS, VENTS, DRAINS, FLANGES, FITTINGS, VALVES, TANKS, PUMP CASINGS AND OTHER PARTS OF THE SYSTEM SUBJECT TO HEAT GAIN OR LOSS AND TO PREVENT CONDENSATION.

E. ALL EQUIPMENT, FITTINGS, DEVICES, ETC REQUIRING SERVICING OR INSPECTION SHALL HAVE REMOVABLE INSULATION WHICH CAN BE REPLACED WITHOUT DAMAGE.

F. ALL LEAK AND PRESSURE TESTS SHALL BE COMPLETED PRIOR TO THE INSTALLATION OF ANY INSULATION.

G. DUCTWORK INSULATION:

1. ALL NEW AND EXISTING SHEET METAL DUCTWORK SHALL BE INSULATED WITH FLEXIBLE DUCT WRAP INSULATION, OF REQUIRED THICKNESS AND DENSITY TO ACHIEVE A MINIMUM INSTALLED R-6 INSULATIVE VALUE AT 75 DEGREES F MEAN TEMPERATURE WHEN LOCATED WITH A CONDITIONED SPACE AND A MINIMUM INSTALLED R-8 INSULATION VALUE WHEN LOCATED IN AN UNCONDITIONED SPACE. INSULATION TO BE PROVIDED WITH REINFORCED FOIL FACED, FLAME RESISTANT, ALUMINUM FOIL VAPOR BARRIER. ALL INSULATION SHALL BE SECURED WITH DUCT ADHESIVE AND SEAMS SEALED BY TWO-INCH SEALING LIP WITH ADHESIVE AND FASTENED WITH 16 GAUGE RUST RESISTANT WIRE OR FIBERGLASS CORD ON 12" CENTERS. ON DUCTS OVER 24" WIDE, WELDED PINS AND CLIPS SHALL BE USED ON THE UNDERSIDE FOR FASTENING INSULATION.

H. PIPING INSULATION:

1. CONDENSATE DRAIN AND DOMESTIC WATER MAKE-UP PIPING SHALL BE INSULATED WITH 1" THICK MOLDED GLASS FIBER WITH A MAXIMUM K FACTOR OF 0.27 AT 75 DECREE F MEAN TEMPERATURE AND FACTORY APPLIED VAPOR BARRIER JACKET.
2. ALL CONDENSER WATER PIPING SHALL BE INSULATED WITH 1-1/2" THICK MOLDED GLASS FIBER FOR PIPE SIZES UP TO 1-1/2" INCHES IN DIAMETER AND 1-1/2" THICK FOR PIPE SIZES LARGER THAN 1-1/2" INCHES IN DIAMETER. INSULATION SHALL HAVE A MAXIMUM K FACTOR OF 0.27 AT 75 DEGREE F MEAN TEMPERATURE AND FACTORY APPLIED VAPOR BARRIER JACKET.
3. HOT WATER PIPING UP TO 220 DEGREES F SHALL BE INSULATED WITH 1-1/2" THICK MOLDED GLASS FIBER FOR PIPE SIZES UP TO 1-1/2" INCHES IN DIAMETER AND 2" THICK FOR PIPE SIZES LARGER THAN 1-1/2" INCHES IN DIAMETER. INSULATION SHALL HAVE A MAXIMUM K FACTOR OF 0.27AT 75 DEGREE F MEAN TEMPERATURE AND FACTORY APPLIED VAPOR BARRIER JACKET.
4. ALL PIPING INSULATION TO BE INSTALLED WITH LONGITUDINAL LAP AND VAPOR BARRIER JOINT SEAL STRIPS WITH ADHESIVE OR SELF-SEALING LAPS. FITTINGS, FLANGES, AND VALVES SHALL BE INSULATED WITH PRE-MOLDED AND PRE-CUT FITTINGS WITH METERED SEGMENTS.

2.05 ACOUSTICAL TREATMENT

A. ACOUSTICAL LINING SHALL MEET THE MINIMUM THERMAL INSULATION VALUE OF R-6 OR A MAXIMUM K FACTOR OF 0.24 AT 1.5" THICKNESS WITH A MEAN TEMPERATURE OF 75 DEGREE F.

B. INSTALL LINER IN ACCORDANCE WITH MANUFACTURERS RECOMMENDATIONS. COMPLETELY COVER ALL PORTIONS OF DUCTWORK PLENUMS AND CASINGS WITH APPROVED ADHESIVE. INSTALL LINER WITH ALL TRAVERSE JOINTS NEATLY BUTTED WITH NO INTERRUPTIONS OR GAPS. COVER ALL EXPOSED EDGES, JOINTS, MECHANICAL FASTENERS AND ANY DAMAGED AREAS WITH ADHESIVE. PROVIDE METAL NOSING AT EQUIPMENT DISCHARGES AND AT END EDGES OF LINING. SECURE LINER WITH APPROVED MECHANICAL FASTENERS INSTALLED IN ACCORDANCE WITH SMACNA DUCT LINER APPLICATION STANDARD.

C. DO NOT EXTERNALLY INSULATE ACOUSTICALLY LINED DUCTS.

D. DO NOT INTERNALLY LINE DUCTWORK WHICH IS A PART OF AN OUTSIDE AIR SYSTEM WHICH DISTRIBUTES UNCONDITIONED AIR.

E. FURNISH AND INSTALL ACOUSTICAL LINING IN DUCTWORK, PLENUMS AND CASINGS AS SHOWN ON THE DRAWINGS AND AS SPECIFIED BELOW:

1. ALL DUCTS WITH DUCT VELOCITIES GREATER THAN 2,000 FPM SHALL HAVE ACOUSTICAL LINING FACED WITH 24 GAUGE PERFORATED ALUMINUM OR GALVANIZED STEEL SUPPORTED 12" ON CENTER.
2. A MINIMUM DISTANCE OF 20 FEET FROM ALL AIR CONDITIONING UNIT INLETS AND DISCHARGES.
3. ALL TERMINAL BOXES SUPPLY DUCTWORK SHALL BE LINED FOR A MINIMUM DISTANCE OF 15 FEET DOWNSTREAM OF BOX DISCHARGE.
4. ALL RETURN/EXHAUST FANS SHALL BE ACOUSTICALLY LINED FOR A MINIMUM DISTANCE OF 20 FEET OF THE FAN INTAKE AND DISCHARGE OPENING.
5. ALL TRANSFER DUCTS SHALL BE PROVIDED WITH 1" THICK ACOUSTICAL LINING FOR ACOUSTICAL PURPOSES ONLY.
6. ALL DUCTWORK PASSING THROUGH OR SERVING CONFERENCE AND MEETING ROOMS SHALL BE PROVIDED WITH ACOUSTICAL LINING.
7. ALL EXPOSED DUCTWORK.

2.06 VIBRATION ISOLATION SYSTEMS

A. ALL ROTATING, REVOLVING OR RECIPROCATING EQUIPMENT, INCLUDING PIPING CONNECTIONS TO THIS EQUIPMENT SHALL BE ACOUSTICALLY ISOLATED TO PREVENT THE TRANSMISSION OF OBJECTIONABLE NOISES, SOUND OR VIBRATIONS TO THE OCCUPIED SPACES AND TO THE BUILDING STRUCTURES. ALL VIBRATION ISOLATION PRODUCTS SHALL BE SPECIFICALLY DESIGNED FOR THEIR INTENDED USE.

B. STATIC DEFLECTION OF ISOLATORS SHALL BE A MINIMUM OF 90% EFFICIENT.

C. MANUFACTURER OF VIBRATION ISOLATION EQUIPMENT SHALL DETERMINE VIBRATION ISOLATOR SIZES AND LOCATIONS, PROVIDE SUITABLE PIPING AND EQUIPMENT VIBRATION ISOLATION SYSTEMS, GUARANTEE SPECIFIED ISOLATION SYSTEM ATTENUATION AND DEFLECTION, AND PROVIDE INSTALLATION INSTRUCTIONS, DRAWINGS AND FIELD SUPERVISION TO ASSURE PROPER INSTALLATION AND PERFORMANCE.

D. MOUNTING TYPES:

1. PROVIDE SPRING ISOLATORS TYPE 30N FOR CEILING-SUPPORTED FANS, IN-LINE PUMPS, HEAT EXCHANGERS, AND AIR HANDLING UNITS. PROVIDE 1" MINIMUM STATIC DEFLECTION.
2. PACKAGED AIR CONDITIONING UNITS WITH INTERNAL ISOLATION OF COMPRESSORS, PROVIDE EXTERNAL SPRINGS (TYPE SLR RESTRAINED FOR FLOOR MOUNTING AND 30N FOR CEILING MOUNTED) WITH 1" MINIMUM STATIC DEFLECTION.
3. PROVIDE FLEXIBLE CONNECTIONS BETWEEN ALL FANS, AHU, AC UNITS AND DUCTWORK AS PER DUCTWORK SPECIFICATION SECTION.

E. FLEXIBLE HOSE CONNECTORS SHALL BE INSTALLED AT INLET AND DISCHARGE CONNECTIONS TO ALL PUMPS.

F. SPRING TYPE 30N HANGERS SHALL BE PROVIDED FOR PIPING FOR A DISTANCE OF 50 FEET OR 50 PIPE DIAMETERS, WHICHEVER IS GREATER, UP AND DOWNSTREAM OF ALL POWER DRIVEN EQUIPMENT. THE HANGER SHALL PROVIDE 1" OF STATIC DEFLECTION FOR PIPES 4" OF OUTSIDE DIAMETER AND LARGER AND 1/2" STATIC DEFLECTION FOR PIPES SMALLER THEN 4" OUTSIDE DIAMETER.

G. VIBRATION ISOLATORS FOR FLOOR OR CEILING SUPPORTED EQUIPMENT SHALL HAVE A MAXIMUM LATERAL MOTION UNDER EQUIPMENT START-UP OR SHUT-DOWN CONDITIONS OF 1/4" AND MOTIONS IN EXCESS SHALL BE RESTRAINED BY SPRING TYPE MOUNTINGS.

H. VIBRATION ISOLATOR SHALL BE PROVIDED BY MASON INDUSTRIES, VIBRATION ELIMINATOR CO., CONSOLIDATED KINETICS CO., OR APPROVED EQUAL.

2.07 EQUIPMENT

A. SERIES FAN POWERED VARIABLE AIR VOLUME (VAV) BOXES:

I. FURNISH AND INSTALL SINGLE DUCT, VARIABLE AIR VOLUME TERMINALS OF THE SIZES AND CAPACITIES SCHEDULED ON THE PLANS. UNITS TO BE MANUFACTURED BY TITUS OR APPROVED EQUAL.

II. SOUND PERFORMANCE SHALL BE ARI CERTIFIED.

III. THE TERMINAL SHALL BE DESIGNED, BUILT, AND TESTED AS A SINGLE UNIT INCLUDING MOTOR AND FAN ASSEMBLY, PRIMARY AIR DAMPER ASSEMBLY, WATER OR ELECTRIC HEATING COILS, AND ACCESSORIES AS SHIPPED. UNIT SHALL SHIP AS A COMPLETE ASSEMBLY REQUIRING NO FIELD ASSEMBLY (INCLUDING ACCESSORIES). ALL ELECTRICAL COMPONENTS SHALL BE UL LISTED AND INSTALLED IN ACCORDANCE WITH THE UL STANDARD 1995. ELECTRICAL CONNECTION SHALL BE SINGLE POINT. ALL ELECTRICAL COMPONENTS, INCLUDING LOW VOLTAGE CONTROLS, SHALL BE MOUNTED IN SHEET METAL CONTROL ENCLOSURES. THE ENTIRE TERMINAL SHALL BE ETL LISTED AS A COMPLETE ASSEMBLY.

IV. THE TERMINAL CASING SHALL BE MINIMUM 20 GAUGE GALVANIZED STEEL. THE UNITS SHALL BE LINED WITH ½-INCH THICK INSULATION, MEETING UL 181 AND NFPA 90A, ENCLOSED BETWEEN THE UNIT CASING AND A NON-PERFORATED INTERNAL 22-GAUGE SHEET METAL COVER EXTENDING OVER THE FIBERGLASS INSULATION, AS WELL AS COVERING THE LINER CUT EDGES. THE DISCHARGE CONNECTION SHALL BE SLIP AND DRIVE CONSTRUCTION FOR ATTACHMENT TO METAL DUCTWORK. THE CASING SHALL BE DESIGNED FOR HANGING BY METAL STRAPS. THE TERMINAL CASING SHALL HAVE A BOTTOM ACCESS PANEL WHICH ALLOWS REMOVAL OF FAN AND SERVICING OF TERMINAL WITHOUT DISTURBING DUCT CONNECTIONS.

V. FAN MOTOR ASSEMBLY SHALL BE FORWARD CURVED CENTRIFUGAL FAN WITH A DIRECT DRIVE MOTOR. MOTORS SHALL BE GENERAL ELECTRIC ECM VARIABLE-SPEED DC BRUSHLESS MOTORS SPECIFICALLY DESIGNED FOR USE WITH SINGLE PHASE, 277 VOLT, 60 HERTZ ELECTRICAL INPUT. MOTOR SHALL BE COMPLETE AND OPERATED BY A SINGLE PHASE INTEGRATED CONTROLLER/INVERTER THAT OPERATES THE WOUND STATOR AND SENSES ROTOR POSITION TO ELECTRONICALLY COMMUTATE THE STATOR. ALL MOTORS SHALL BE DESIGNED FOR SYNCHRONOUS ROTATION. ROTOR SHALL BE PERMANENT MAGNET TYPE WITH NEAR ZERO ROTOR LOSSES. MOTOR SHALL HAVE BUILT-IN SOFT START AND SOFT SPEED CHANGE RAMPS. MOTOR SHALL BE ABLE TO BE MOUNTED WITH SHAFT IN HORIZONTAL OR VERTICAL ORIENTATION. MOTOR SHALL BE PERMANENTLY LUBRICATED WITH BALL BEARINGS. MOTOR SHALL BE DIRECTLY COUPLED TO THE BLOWER. MOTOR SHALL MAINTAIN A MINIMUM OF 70 PERCENT EFFICIENCY OVER ITS ENTIRE OPERATING RANGE. PROVIDE A MOTOR THAT IS DESIGNED TO OVERCOME REVERSE ROTATION AND NOT AFFECT LIFE EXPECTANCY.

VI. THE TERMINALS SHALL UTILIZE A MANUAL SCR, WHICH ALLOWS CONTINUOUSLY ADJUSTABLE FAN SPEED FROM MAXIMUM TO MINIMUM, AS A MEANS OF SETTING FAN AIRFLOW. SETTING FAN AIRFLOW WITH ANY DEVICE THAT RAISES THE PRESSURE ACROSS THE FAN TO REDUCE AIRFLOW IS NOT ACCEPTABLE. THE SPEED CONTROL SHALL INCORPORATE A MINIMUM VOLTAGE STOP TO ENSURE THAT THE MOTOR CANNOT OPERATE IN A STALL MODE.

VII. THE PRIMARY AIR DAMPER ASSEMBLY SHALL BE HEAVY GAUGE STEEL WITH SHAFT ROTATING IN DELRIN BEARINGS. NYLON BEARINGS ARE NOT ACCEPTABLE. DAMPER LEAKAGE SHALL NOT EXCEED 5 PERCENT OF THE MANUFACTURERS' SCHEDULED MAXIMUM FAN CAPACITY AT 1 INCH WG. INLET STATIC PRESSURE.

VIII. HOT WATER HEATING COILS SHALL BE ENCLOSED IN A MINIMUM 20-GAUGE GALVANIZED STEEL CASING, WITH FLANGED CONSTRUCTION FOR ATTACHMENT TO METAL DUCTWORK. COILS SHALL BE FACTORY INSTALLED ON THE TERMINAL. FINS SHALL BE RIPPLED AND CORRUGATED HEAVY GAUGE ALUMINUM, MECHANICALLY BONDED TO TUBES. TUBES SHALL BE COPPER WITH MINIMUM WALL THICKNESS OF 0.016 INCH, WITH MALE SOLDER HEADER CONNECTIONS. COILS SHALL BE LEAK TESTED TO 300 PSI, WITH MINIMUM BURST PRESSURE OF 1800 PSI AT AMBIENT TEMPERATURE. NUMBER OF COIL ROWS AND CIRCUITS SHALL BE SELECTED TO PROVIDE PERFORMANCE AS REQUIRED PER THE PLANS. COIL PERFORMANCE DATA SHALL BE BASED ON TESTS RUN IN ACCORDANCE WITH ARI STANDARD 410.

IX. AT AN INLET VELOCITY OF 2000 FPM, THE DIFFERENTIAL STATIC PRESSURE REQUIRED TO OPERATE ANY TERMINAL SIZE SHALL NOT EXCEED 0.18" WG.

X. MECHANICAL CONTRACTOR TO COORDINATE THE PROVISION OF ALL POWER, CONTROL WIRING, TRANSFORMERS, DISCONNECT SWITCHES, ETC. WITH A.T.C. CONTRACTOR AS REQUIRED.

XI. DDC CONTROLS: DDC CONTROLLER, ACTUATOR, TEMPERATURE SENSORS SHALL BE SUPPLIED BY ATC. CONTROLLER, ACTUATOR SHALL BE SHIPPED TO VAV MANUFACTURER'S FACTORY FOR MOUNTING. ALL CONTROLS SHALL BE MOUNTED WITHIN STEEL CONTROL ENCLOSURE. DDC CONTROL SHALL BE FIELD CALIBRATED BY ATC.

B. PLENUM RATED CONDENSATE PUMPS (REQUIRED FOR ALL ABOVE CEILING APPLICATIONS):

I. FURNISH AND INSTALL CONDENSATE PUMPS AND CONNECT TO DRAINS AS INDICATED ON THE DRAWINGS.

II. CONDENSATE PUMP SHALL BE PLENUM RATED IN-LINE, AUTOMATIC PUMP SIMILAR TO HARTELL TYPE A2X-1965 DV. PUMP SHALL BE CAPABLE OF OPERATION WITH 120V, 1 PHASE, 60 HZ POWER. PUMP SHALL BE RATED AT 240 GPH @ 12 FEET HEAD.

III. PUMP SHALL HAVE 1/10 HP MOTOR WITH SINGLE POINT, HARD WIRED ELECTRICAL CONNECTION.

IV. CONDENSATE PUMP SHALL HAVE FOLLOWING FEATURES: VERTICAL TYPE PUMP UNIT; LEAKPROOF, RUSTPROOF, HEAVY DUTY CAST ALUMINUM CONSTRUCTION; DUAL VOLTAGE CAPABLE; SNAP ACTION SWITCH; AUXILARY SAFETY SWITCH; THERMAL OVERLOAD PROTECTION.

V. PROVIDE DISCONNECT SWITCH AND CHECK VALVE AT PUMP DISCHARGE.

C. IN-LINE CABINET EXHAUST/TRANSFER FANS

I. CEILING MOUNTED EXHAUST FANS SHALL BE OF THE CENTRIFUGAL DIRECT DRIVE TYPE.

II. THE FAN HOUSING SHALL BE CONSTRUCTED OF HEAVY GAUGE GALVANIZED STEEL. THE HOUSING INTERIOR SHALL BE LINED WITH 1/2" ACOUSTICAL INSULATION.

III. THE OUTLET DUCT COLLAR SHALL INCLUDE AN ALUMINUM BACKDRAFT DAMPER AND SHALL BE ADAPTABLE FOR HORIZONTAL OR VERTICAL DISCHARGE.

IV. THE ACCESS FOR WIRING SHALL BE EXTERNAL. THE MOTOR DISCONNECT SHALL BE INTERNAL AND OF THE PLUG-IN TYPE. THE MOTOR SHALL BE MOUNTED ON VIBRATION ISOLATORS.

V. THE FAN WHEEL(S) SHALL BE OF THE FORWARD CURVED CENTRIFUGAL TYPE AND DYNAMICALLY BALANCED.

VI. ALL FANS SHALL BEAR THE AMCA CERTIFIED RATINGS SEAL FOR SOUND AND AIR PERFORMANCES AND SHALL BE U.L. LISTED AND C.S.A. APPROVED.

VII. A MANUFACTURER SUPPLIED SPEED SWITCH SHALL BE PROVIDED WITH EACH EXHAUST FAN TO BE MOUNTED ON THE HOUSING FOR BALANCING PURPOSES.

VIII. FANS SHALL BE MANUFACTURED BY GREENHECK OR EQUAL.

D. VERTICAL IN-LINE MULTI-STAGE PUMPS

I. THE PUMPS SHALL BE ANSI/NSF 61 APPROVED FOR DRINKING WATER.

II. THE PUMPS SHALL BE OF THE IN-LINE VERTICAL MULTI-STAGE DESIGN

III. THE HEAD-CAPACITY CURVE SHALL HAVE A STEADY RISE IN HEAD FROM MAXIMUM TO MINIMUM FLOW WITHIN THE PREFERRED OPERATING REGION. THE SHUT-OFF HEAD SHALL BE A MINIMUM OF 20% HIGHER THAN THE HEAD AT THE BEST EFFICIENCY POINT.

IV. SMALL VERTICAL IN-LINE MULTI-STAGE PUMPS (12MM OR 16MM SHAFT, NOMINAL FLOW FROM 3 TO 125 GALLONS PER MINUTE) SHALL HAVE THE FOLLOWING FEATURES:

1. THE PUMP IMPELLERS SHALL BE SECURED DIRECTLY TO THE PUMP SHAFT BY MEANS OF A SPLINED SHAFT ARRANGEMENT.
2. THE SUCTION/DISCHARGE BASE SHALL HAVE ANSI CLASS 250 FLANGE OR INTERNAL PIPE THREAD (NPT) CONNECTIONS AS DETERMINED BY THE PUMP STATION MANUFACTURER.
3. PUMP CONSTRUCTION.
   - 1. SUCTION/DISCHARGE BASE, PUMP HEAD, MOTOR STOOL:CAST IRON (CLASS 30)
   - 2. IMPELLERS, DIFFUSER CHAMBERS, OUTER SLEEVE:304 STAINLESS STEEL
   - 3. SHAFT 316 OR 431 STAINLESS STEEL
   - 4. IMPELLER WEAR RINGS:304 STAINLESS STEEL
   - 5. SHAFT JOURNALS AND CHAMBER BEARINGS: SILICON CARBIDE
   - 6. O-RINGS:EPDM
   - 7. SHAFT COUPLINGS SHALL BE MADE OF CAST IRON OR SINTERED STEEL.
4. THE SHAFT SEAL SHALL BE A BALANCED O-RING CARTRIDGE TYPE WITH THE FOLLOWING FEATURES:
   - 1. COLLAR, DRIVERS, SPRING:316 STAINLESS STEEL
   - 2. SHAFT SLEEVE, GLAND PLATE:316 STAINLESS STEEL
   - 3. STATIONARY RING: SILICON CARBIDE (GRAPHITE IMBEDDED)
   - 4. ROTATING RING: SILICON CARBIDE (GRAPHITE IMBEDDED)
   - 5. O-RINGS: EPDM
   - 6. SHAFT SEAL REPLACEMENT SHALL BE POSSIBLE WITHOUT REMOVAL OF ANY PUMP COMPONENTS OTHER THAN THE COUPLING GUARD, SHAFT COUPLING AND MOTOR.
5. EACH MOTOR SHALL BE OF THE INTEGRATED VARIABLE FREQUENCY DRIVE DESIGN CONSISTING OF A MOTOR AND A VARIABLE FREQUENCY DRIVE (VFD) WITH A BUILT-IN PUMP SYSTEM CONTROLLER. THE COMPLETE VFD/MOTOR ASSEMBLY SHALL BE BUILT AND TESTED AS ONE UNIT BY THE SAME MANUFACTURER.
6. THE VFD/MOTOR SHALL HAVE AN IP55 (TEFC) ENCLOSURE RATING AS A COMPLETE ASSEMBLY. THE MOTOR SHALL HAVE A STANDARD NEMA C-FACE, CLASS F INSULATION WITH A CLASS B TEMPERATURE RISE.
7. THE VFD SHALL BE OF THE PWM (PULSE WIDTH MODULATION) DESIGN USING UP TO DATE IGBT (INSULATED GATE BIPOLAR TRANSISTOR) TECHNOLOGY.
8. THE VFD SHALL CONVERT INCOMING FIXED FREQUENCY THREE-PHASE AC POWER INTO A VARIABLE FREQUENCY AND VOLTAGE FOR CONTROLLING THE SPEED OF THE MOTOR. THE MOTOR CURRENT SHALL CLOSELY APPROXIMATE A SINE WAVE. MOTOR VOLTAGE SHALL BE VARIED WITH FREQUENCY TO MAINTAIN DESIRED MOTOR MAGNETIZATION CURRENT SUITABLE FOR CENTRIFUGAL PUMP CONTROL AND TO ELIMINATE THE NEED FOR MOTOR DE-RATING.
9. THE VFD SHALL HAVE, AS A STANDARD COMPONENT, AN RFI FILTER (RADIO FREQUENCY INTERFERENCE) TO MINIMIZE ELECTRICAL NOISE DISTURBANCES BETWEEN THE POWER ELECTRONICS AND THE POWER SUPPLY. THE VFD/MOTOR SHALL MEET ALL REQUIREMENTS OF THE EMC DIRECTIVE CONCERNING RESIDENTIAL AND LIGHT INDUSTRY EQUIPMENT (EN 61800-3).
10. THE VFD SHALL HAVE A MINIMUM OF TWO SKIP FREQUENCY BANDS WHICH CAN BE FIELD ADJUSTABLE.
11. THE VFD SHALL HAVE INTERNAL SOLID-STATE OVERLOAD PROTECTION DESIGNED TO TRIP WITHIN THE RANGE OF 125-150% OF RATED CURRENT.
12. THE VFD/MOTOR SHALL INCLUDE PROTECTION AGAINST INPUT TRANSIENTS, LOSS OF AC LINE PHASE, OVER-VOLTAGE, UNDER-VOLTAGE, VFD OVER-TEMPERATURE, AND MOTOR OVER-TEMPERATURE. THE MOTOR OVER-TEMPERATURE PROTECTION SHALL CONSIST OF THREE SERIES CONNECTED PTC THERMISTORS, ONE FOR EACH MOTOR PHASE.
13. THE VFD/MOTOR SHALL PROVIDE FULL NAMEPLATE OUTPUT CAPACITY (HORSEPOWER AND SPEED) WITHIN A BALANCED VOLTAGE RANGE OF 414 TO 528 VOLTS.
14. AUTOMATIC DE-RATE FUNCTION: THE VFD/MOTOR SHALL REDUCE SPEED DURING PERIODS OF OVERLOAD ALLOWING FOR REDUCED CAPACITY PUMP OPERATION WITHOUT COMPLETE SHUT-DOWN OF THE SYSTEM. DETECTION OF OVERLOAD SHALL BE BASED ON CONTINUOUS MONITORING OF CURRENT, VOLTAGE AND TEMPERATURE WITHIN THE VFD/MOTOR ASSEMBLY.
15. THE VFD/MOTOR SHALL HAVE, AS A MINIMUM, THE FOLLOWING INPUT/OUTPUT CAPABILITIES:
    - 1. SPEED REFERENCE SIGNAL: 0-10 VDC, 4-20MA
    - 2. DIGITAL REMOTE ON/OFF
    - 3. FAULT SIGNAL RELAY (NC OR NO)
    - 4. FIELDBUS COMMUNICATION PORT (RS485)
16. MOTOR DRIVE END BEARINGS SHALL BE ADEQUATELY SIZED SO THAT THE MINIMUM L10 BEARING LIFE IS 17,500 HOURS AT THE MINIMUM ALLOWABLE CONTINUOUS FLOW RATE FOR THE PUMP AT FULL RATED SPEED.
17. THE PUMP SYSTEM CONTROLLER (PROPORTIONAL-INTEGRAL) SHALL BE A STANDARD COMPONENT OF THE INTEGRATED VARIABLE FREQUENCY DRIVE MOTOR DEVELOPED AND SUPPORTED BY THE PUMP MANUFACTURER.
18. THE PUMP SYSTEM CONTROLLER SHALL HAVE AN EASY TO USE INTERFACE MOUNTED ON THE VFD/MOTOR ENCLOSURE. PUMP SYSTEM START/STOP AND SET-POINT ADJUSTMENT SHALL BE POSSIBLE THROUGH THE USE OF TWO PUSH BUTTONS LOCATED ON THE DRIVE ENCLOSURE.
19. THE VFD/MOTOR SHALL BE CAPABLE OF RECEIVING A REMOTE ANALOG SET-POINT (4-20MA OR 0-10 VDC) AS WELL AS A REMOTE ON/OFF (DIGITAL) SIGNAL.
20. PUMP STATUS AND ALARM STATE SHALL BE INDICATED VIA TWO LED LIGHTS LOCATED ON THE VFD/MOTOR ENCLOSURE.
21. ADVANCED PROGRAMMING AND TROUBLESHOOTING SHALL BE POSSIBLE VIA AN INFRA-RED HAND HELD PROGRAMMER OR A FIELD CONNECTED PERSONAL COMPUTER. PUMP SYSTEM PROGRAMMING (FIELD ADJUSTABLE) SHALL INCLUDE AS A MINIMUM THE FOLLOWING:
    - 1. SYSTEM PRESSURE SET-POINT, PSIG
    - 2. SYSTEM START PRESSURE, PSIG
    - 3. SYSTEM STOP PRESSURE, PSIG
    - 4. MINIMUM PUMP SPEED, %
    - 5. PRESSURE TRANSDUCER SUPPLY/RANGE
    - 6. MAXIMUM PUMP SPEED, %
    - 7. SYSTEM TIME (PROPORTIONAL GAIN)
    - 8. INTEGRAL ACTION TIME
22. THE INFRA-RED PROGRAMMER SHALL BE CAPABLE OF DISPLAYING THE FOLLOWING STATUS READINGS:
    - 1. PUMP STATUS (ON, OFF, MIN., MAX.)
    - 2. SYSTEM SET-POINT, PSIG
    - 3. ACTUAL SYSTEM PRESSURE, PSIG
    - 4. REMOTE SET-POINT, %
    - 5. PUMP SPEED, RPM
    - 6. VFD/MOTOR INPUT POWER, KW
    - 7. VFD/MOTOR TOTAL CUMULATIVE KWH
    - 8. VFD/MOTOR TOTAL OPERATING HOURS
23. THE INFRA-RED PROGRAMMER SHALL ALSO BE CAPABLE OF DISPLAYING THE FOLLOWING ALARMS, WITH THE LAST FIVE ALARMS STORED IN MEMORY:
    - 1. LOSS OF SENSOR SIGNAL
    - 2. LOSS OF EXTERNAL SET-POINT SIGNAL
    - 3. UNDER-VOLTAGE & OVER-VOLTAGE
    - 4. MOTOR OVERLOAD (BLOCKED PUMP)
    - 5. MOTOR OVER-TEMPERATURE
    - 6. DRIVE OVER-TEMPERATURE
    - 7. DRIVE OVER-CURRENT
24. THE CRE PUMP SHALL HAVE A MAXIMUM WORKING PRESSURE RANGING UP TO 435 PSIG AT A TEMPERATURE OF 176°F.
25. A DIAPHRAGM TANK (FIELD INSTALLED) IS REQUIRED FOR THE STOP FUNCTION. A MINIMUM DIAPHRAGM TANK CONNECTION SIZE OF ¾" (THREE QUARTER INCH) SHALL BE PROVIDED ON THE DISCHARGE PIPING BY OTHERS.
26. A PRESSURE TRANSDUCER SHALL BE FACTORY INSTALLED ON THE PUMP HEAD. PRESSURE TRANSDUCERS SHALL BE MADE OF 316 STAINLESS STEEL. TRANSDUCER ACCURACY SHALL BE +/- 1.0% FULL SCALE WITH HYSTERESIS AND REPEATABILITY OF NO GREATER THAN 0.1% FULL SCALE. THE OUTPUT SIGNAL SHALL BE 4-20 MA WITH A SUPPLY VOLTAGE RANGE OF 9-32 VDC.
27. A FACTORY PRESSURE SWITCH SHALL BE MOUNTED ON THE SUCTION PIPING FOR WATER SHORTAGE PROTECTION. THE PUMP SHUT-DOWN PRESSURE SHALL BE 3 PSIG WITH A RESET PRESSURE OF 5 PSIG. SYSTEMS WITH A FLOODED SUCTION INLET OR SUCTION LIFT CONFIGURATION WILL REQUIRE A FIELD INSTALLED WATER SHORTAGE DEVICE. A NORMALLY OPEN DRY CONTACT SHALL BE AVAILABLE ON THE VFD/MOTOR FOR FIELD INSTALLATION.
28. THE ENTIRE PUMP MOTOR AND VFD SHALL BE FACTORY PERFORMANCE TESTED AS A COMPLETE UNIT PRIOR TO SHIPMENT.
29. THE PUMP SHALL UNDERGO A HYDROSTATIC TEST.
30. THE WARRANTY PERIOD SHALL BE A NON-PRORATED PERIOD OF 24 MONTHS FROM DATE OF INSTALLATION, NOT TO EXCEED 30 MONTHS FROM DATE OF MANUFACTURE.

E. ABOVE CEILING MOUNTED WATER COOLED SUPPLAMENTAL AC UNIT:

I. UNITS SHALL BE PERFORMANCE CERTIFIED TO ISO STANDARD 13256-1 FOR WATER LOOP HEAT PUMP. UNIT SHALL BE UNDERWRITER LABORATORIES (UL) LISTED FOR SAFETY ON ALL MODELS. EACH UNIT SHALL BE RUN TESTED AT THE FACTORY. EACH UNIT SHALL BE PALLET MOUNTED AND STRETCH WRAPPED. THE UNITS SHALL BE DESIGNED TO OPERATE WITH ENTERING FLUID TEMPERATURES BETWEEN 50 DEGREES F AND 110 DEGREES F IN COOLING AND 25 DEGREES F AND 80 DEGREES F IN HEATING.

II. THE UNIT SHALL BE WARRANTED BY THE MANUFACTURER AGAINST DEFECTS IN MATERIALS AND WORKMANSHIP FOR A PERIOD OF FIVE YEARS ON THE COMPRESSOR AND ONE YEAR ON ALL OTHER PARTS.

III. THE CABINET SHALL BE FABRICATED FROM HEAVY-GAUGE G-90 GALVANIZED STEEL FINISHED WITH GALVALUME PLUS. HANGING BRACKETS SHALL BE PROVIDED AS STANDARD FOR HORIZONTAL UNITS. THE INTERIOR SHALL BE INSULATED WITH ½" THICK, MULTI DENSITY, COATED GLASS FIBER. INSULATION WITHIN THE AIR HANDLING SECTION SHALL NOT HAVE ANY EXPOSED EDGES. ALL INSULATION MUST MEET NFPA 90A AND BE CERTIFIED TO MEET THE GREENGUARD® INDOOR AIR QUALITY STANDARD FOR LOW EMITTING PRODUCTS. ALL UNITS SHALL ALLOW SUFFICIENT SERVICE ACCESS TO REPLACE THE COMPRESSOR WITHOUT UNIT REMOVAL. ONE BLOWER AND TWO COMPRESSOR COMPARTMENT ACCESS PANELS SHALL BE REMOVABLE WITH SUPPLY AND RETURN DUCTWORK IN PLACE. A DUCT COLLAR SHALL BE PROVIDED ON THE SUPPLY AIR OPENING. A 2" RETURN AIR FILTER RACK/DUCT COLLAR WITH 2" THICK MERV 8 FILTERS SHALL BE PROVIDED WITH EACH UNIT. THE UNITS SHALL HAVE AN INSULATED DIVIDER PANEL BETWEEN THE AIR HANDLING SECTION AND THE COMPRESSOR SECTION TO MINIMIZE THE TRANSMISSION OF COMPRESSOR NOISE, AND TO PERMIT SERVICE TESTING WITHOUT AIR BYPASS. UNITS SHALL BE PROVIDED WITH A STAINLESS STEEL CONDENSATE DRAIN PAN.

IV. REFRIGERATION CIRCUITS: ALL UNITS SHALL CONTAIN A SEALED REFRIGERANT CIRCUIT INCLUDING A HERMETICTIC TWO STAGE COMPRESSOR DESIGNED FOR R-410A, BI-DIRECTIONAL THERMAL EXPANSION VALVE METERING DEVICE, FINNED TUBE AIR-TO-REFRIGERANT HEAT EXCHANGER, AND SERVICE PORTS. COMPRESSOR SHALL BE HIGH EFFICIENCY SCROLL TYPE, QUIET OPERATION AND MOUNTED ON RUBBER VIBRATION ISOLATORS. UNIT SHALL HAVE A FLOATING COMPRESSOR OR PAN CONSISTING OF A 1/2" THICK HIGH DENSITY ELASTOMERIC PAD BETWEEN THE COMPRESSOR BASE PLATE AND THE UNIT BASE PAN TO PREVENT TRANSMISSION OF VIBRATION TO THE STRUCTURE.

V. COMPRESSOR MOTORS SHALL BE EQUIPPED WITH OVERLOAD PROTECTION. THE FINNED TUBE COIL SHALL BE CONSTRUCTED OF LANCED ALUMINUM FINS NOT EXCEEDING SIXTEEN FINS PER INCH BONDED TO RIFLED COPPER TUBES IN A STAGGERED PATTERN NOT LESS THAN THREE ROWS DEEP AND HAVE A 450 PSIG WORKING PRESSURE. COILS SHALL HAVE A BAKED POLYESTER ENAMEL COATING FOR PROTECTION AGAINST AIRBORNE CHEMICALS. COIL END PLATES SHALL BE ALUMINUM. THE COAXIAL WATER-TO-REFRIGERANT HEAT EXCHANGER SHALL BE CONSTRUCTED OF A CONVOLUTED COPPER INNER TUBE AND STEEL OUTER TUBE WITH A DESIGNED REFRIGERANT WORKING PRESSURE OF 450 PSIG AND A DESIGN WATER SIDE WORKING PRESSURE OF NO LESS THAN 400 PSIG. THE WATER-TO-REFRIGERANT HEAT EXCHANGER SHALL BE INSULATED TO PREVENT CONDENSATION AT LOW FLUID TEMPERATURES.

VI. FAN MOTOR & ASSEMBLY: THE FAN SHALL BE DIRECT DRIVE CENTRIFUGAL FORWARD CURVED TYPE WITH A DYNAMICALLY

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NO.	DATE	ISSUE
3	07/01/2026	ISSUE FOR ADD ALTERNATE
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REVISIONS

PROJECT:
SNOWFLAKE
7 TIMES SQUARE, NEW YORK, NY
FLOORS 29, 30, 31

DRAWING TITLE:
MECHANICAL SPECIFICATIONS SHEET 1 OF 2

SEAL & SIGNATURE

DATE: 05.22.2026
SCALE: 12" = 1'-0"
PROJECT NO. ----
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BALANCED WHEEL. THE HOUSING AND WHEEL SHALL BE DESIGNED FOR QUIET LOW VELOCITY OPERATION. THE FAN HOUSING SHALL BE REMOVABLE FROM THE UNIT WITHOUT DISCONNECTING THE SUPPLY AIR DUCTWORK FOR SERVICING OF THE FAN MOTOR. THE FAN MOTOR SHALL BE AN ECM-2 MICROPROCESSOR CONTROLLED DC TYPE MOTOR WITH INTERNAL PROGRAMMING FACTORY SET FOR THE SPECIFIC UNIT AND FEATURING SOFT START/STOP AND DELAY OFF FEATURE FOR MAXIMUM EFFICIENCY AND QUIET OPERATION. AIR FLOW RATES SHALL BE VARIED ACCORDING TO THE STAGING OF THE UNIT. THERE WILL FURTHER BE PROVISIONS FOR ADJUSTING THE AIR DELIVERY OF THE MOTOR AND BLOWER BY +/- 15% FROM THE RATED AIRFLOW.

VII. ELECTRICAL: CONTROLS AND SAFETY DEVICES WILL BE FACTORY WIRED AND MOUNTED WITHIN THE UNIT. CONTROLS SHALL INCLUDE FAN RELAY, COMPRESSOR CONTACTOR, 24V (50 VA) TRANSFORMER, REVERSING VALVE AND SOLID STATE LOCKOUT CONTROLLER (UPM). THE UPM CONTROLLER SHALL INCLUDE THE FOLLOWING FEATURES:

1. DIAGNOSTIC LED'S
2. LOW PRESSURE BYPASS TIME DELAY
3. ANTI-SHORT CYCLE TIME DELAY
4. RANDOM START TIME DELAY AND ONE TIME INTELLIGENT RESET.

VIII. A LOCK-OUT INDICATING TERMINAL SHALL BE PROVIDED IN THE LOW VOLTAGE CIRCUIT. SAFETY DEVICES INCLUDE A LOW PRESSURE CUTOUT SET AT 40 PSIG FOR LOSS OF CHARGE PROTECTION AND A HIGH PRESSURE CUTOUT CONTROL SET AT 600 PSIG. THE ECM MOTOR INTERFACE BOARD SHALL PROVIDE A SCREW TYPE TERMINAL BLOCK FOR THERMOSTAT CONNECTION, LED'S TO INDICATE THERMOSTAT STATUS AND AIR DELIVERY. IT SHALL ALSO PROVIDE A MEANS OF CHANGING THE MOTOR PROGRAM TO ANY OF UP TO FOUR PRE-PROGRAMMED OPTIONS. DIRECT WIRING OF THE MOTOR CONTROL HARNESS TO THE THERMOSTAT IS UNACCEPTABLE. A TERMINAL BLOCK WITH SCREW TERMINALS SHALL BE PROVIDED FOR CONTROL WIRING. A CONDENSATE OVERFLOW SWITCH SHALL BE FACTORY INSTALLED TO STOP COMPRESSOR OPERATION IF DRAIN PAN OVERFLOW IS EMINENT. AN ENERGY MANAGEMENT RELAY TO ALLOW UNIT CONTROL BY AN EXTERNAL SOURCE SHALL BE FACTORY INSTALLED.

IX. PIPING: SUPPLY, RETURN WATER AND CONDENSATE DRAIN CONNECTIONS SHALL BE BRASS FEMALE PIPE THREAD FITTINGS AND MOUNTED FLUSH TO CABINET EXTERIOR.

X. PROVIDE THE FOLLOWING ACCESSORIES:

1. HOT GAS BYPASS - HOT GAS BYPASS VALVING SHALL BE FACTORY MOUNTED AND INSTALLED INTEGRAL TO THE UNIT. THE HOT GAS BYPASS VALVE SHALL RUN OFF REFRIGERANT PRESSURE AND ALLOW UNIT TO RUN ON LOW LOAD CONDITIONS.
2. EXTRA QUIET SOUND PACKAGE - THE UNIT SHALL BE PROVIDED WITH A FACTORY BUILT AND TESTED EXTRA QUIET SOUND PACKAGE CONSISTING OF ½” INCH HIGH DENSITY CLOSED CELL FOAM INSULATION WRAP ON THE COMPRESSOR AND AN INSULATED DIVIDER PANEL BETWEEN THE AIR- HANDLING SECTION AND THE COMPRESSOR SECTION TO MINIMIZE THE TRANSMISSION OF COMPRESSOR NOISE. ALL BULK HEADS ARE TO BE PROVIDED WITH 1” RUBATEX INSULATION. THE CABINET INTERIOR SHALL BE INSULATED WITH ½” THICK, MULTI-DENSITY COATED GLASS FIBER. THE UNIT SHALL ALSO HAVE A HEAVY GAUGE STEEL COMPRESSOR BASE PAN WITH WELDED HEAVY GAGE MOUNTING RAILS FOR COMPRESSOR MOUNTING. THE COMPRESSORS ARE TO BE MOUNTED ON RESILIENT RUBBER GROMMETS FOR VIBRATION DAMPING.
3. LOW WATER TEMPERATURE PACKAGE FOR UNIT OPERATION DOWN TO 45F EWT IN THE COOLING MODE WITHOUT THE USE OF A WATER REGULATING VALVE CONSISTS OF:

1. AN EXTERNALLY EQUALIZED TXV FOR HEAD PRESSURE CONTROL.
2. WRAPPED CONDENSER.
3. LOW PRESSURE TIME DELAY RELAY

F. CEILING-MOUNTED ODOR CONTROL UNIT:

I. PROVIDE A SELF-CONTAINED, NON-DUCTED AIR PURIFICATION UNIT DESIGNED FOR FLUSH INSTALLATION IN A STANDARD 48" X 24" SUSPENDED “T-GRID” CEILING. UNIT SHALL BE CAPABLE OF DELIVERING 500-600 CFM WITH ALL CLEAN FILTERS INSTALLED AND SHALL INCLUDE AN INTEGRAL FAN ASSEMBLY, MULTI-STAGE FILTRATION, GAS-PHASE ADSORPTION MEDIA, BIPOLAR IONIZATION DEVICE, CONTROLS, AND MONITORING COMPONENTS. UNIT SHALL BE SUITABLE FOR CONTROLLING PARTICULATE MATTER, ODORS, AND VOCS IN OCCUPIED SPACES SUCH AS OFFICES AND FOOD PANTRY AREAS. THE AIR PURIFICATION PROCESS SHALL INCLUDE: A 1-INCH MERV 7 PREFILTER; TWO (2) 2-INCH HONEYCOMB ACTIVATED CARBON PANELS WITH BS-145XL MEDIA FOR GAS-PHASE REMOVAL; A 4-INCH MERV 13 PRIMARY PARTICULATE FILTER; AND ONE (1) BIPOLAR IONIZATION MODULE. UNIT SHALL BE CONSTRUCTED OF 5052 ALUMINUM, INTERNALLY LINED WITH 1-INCH ACOUSTICAL INSULATION, AND SEALED TO PREVENT AIR BYPASS. PROVIDE WIRELESS REMOTE CONTROL, 0-10 VDC BMS FAN SPEED CONTROL, DIRTY FILTER INDICATION WITH DIFFERENTIAL PRESSURE SWITCHES, AND A UNIT-MOUNTED DISCONNECT. ELECTRICAL SERVICE SHALL BE 120 OR 240 VAC, SINGLE PHASE, 60 HZ.

2.08 MOTOR STARTERS, CONTROL DEVICES AND MOTORS

A. MECHANICAL CONTRACTOR TO FURNISH AND INSTALL STARTERS FOR POWER WIRING BY THE ELECTRICAL CONTRACTOR.

B. MOTOR STARTERS SHALL BE CUTLER HAMMER, WESTINGHOUSE OR ALLEN-BRADLEY MANUFACTURER, SUITABLE FOR WALL OR ANGLE IRON FRAME MOUNTING.

C. ALL STARTERS FOR MOTORS LESS THAN 1/2 HP SHALL BE 120 VOLT, SINGLE PHASE, 60 CYCLE, A.C. SERVICE. MANUAL STARTERS WITH OVERLOAD PROTECTION AND LOCKOUT TYPE DISCONNECT SWITCH OR BREAKER MAY BE USED TO CONTROL SUCH MOTORS, EXCEPT WHERE INTERLOCKS OR AUTOMATIC CONTROLS ARE REQUIRED. IN SUCH CASES, MAGNETIC ACROSS-THE-LINE STARTERS SHALL BE FURNISHED.

D. ALL STARTERS FOR MOTORS 1/2 HP TO 75 HP SHALL BE COMBINATION FUSED DISCONNECT, MAGNETIC ACROSS-THE-LINE TYPE WITH FUSIBLE SWITCH. STARTERS 75 HP AND GREATER SHALL BE SOLID STATE ELECTRONIC SOFT START TYPE STARTERS.

E. CONTROLLERS FOR CONDENSATE PUMPS, DUPLEX AIR COMPRESSOR, SUMP AND EJECTOR PUMPS, ETC., SHALL BE FACTORY MOUNTED AND WIRED AS PART OF THE WORK OF THE HEATING, VENTILATING AND AIR CONDITIONING SECTION.

F. ALL MAGNETIC STARTERS SUBJECT TO MANUAL START AND IN DIRECT VIEW OF THE MOTORS THEY CONTROL SHALL HAVE MOMENTARY CONTACT START AND STOP BUTTONS AND PILOT LIGHT BUILT IN TO COVER. ALL MAGNETIC STARTERS SUBJECT TO ELECTRICAL INTERLOCK OR AUTOMATIC CONTROL SHALL HAVE HAND-OFF-AUTOMATIC SWITCHES AND PILOT LIGHT BUILT INTO COVER.

G. WHERE STARTERS ARE NOT IN SIGHT OF MOTORS THEY CONTROL, A LOCAL DISCONNECT SWITCH WILL BE PROVIDED BY THE ELECTRICAL CONTRACTOR.

H. PROVIDE ALL STARTERS WITH TRANSFORMERS BUILT INTO EACH STARTER CASING FOR CONTROL CIRCUIT. TRANSFORMERS SHALL SERVE ALL CONTROL CIRCUITS. EACH STARTER SUBJECT TO ELECTRICAL INTERLOCK AND/OR AUTOMATIC CONTROL SHALL HAVE THE NECESSARY AUXILIARY CONTACTS. ONE SET OF TERMINALS SHALL BE PROVIDED FOR EACH CONTROL CIRCUIT. CONTROL CENTERS SHALL BE PROVIDED WITH CONTROL TERMINAL BLOCKS. PROVIDE THREE SETS OF NORMALLY CLOSED OR NORMALLY OPEN CONTACTS.

I. ALL MAGNETIC STARTERS SHALL HAVE THERMAL OVERLOAD IN EACH PHASE LEG AND LOW VOLTAGE PROTECTION.

J. ALL PARTS SUBJECT TO WEAR, ARCING, ETC., SHALL BE REPLACEABLE.

K. ALL WIRING, STARTERS, SWITCHES, ETC., SHALL BE IN FULL ACCORDANCE WITH ALL LOCAL INSURANCE UNDERWRITERS CODE REQUIREMENTS.

L. FURNISH DETAILED COMPOSITE WIRING DIAGRAMS FOR THOSE INSTALLING ELECTRICAL WORK, AND FURNISH SUCH OTHER INFORMATION NECESSARY TO ASSURE THE PROPER CONNECTION, OPERATION AND CONTROL OF MOTORIZED EQUIPMENT, INCLUDING INTERLOCKS, AUTOMATIC OR SAFETY CONTROLS AND AUXILIARY CIRCUITS.

M. FURNISH THE PERTINENT INFORMATION SUCH AS STARTING TORQUE REQUIREMENTS OF HIGH INERTIA EQUIPMENT, SO THAT THE PROPER TYPE STARTER MAY BE PROVIDED BY THE STARTER MANUFACTURER. ALL INFORMATION IS SUBJECTED TO THE REVIEW OF THE ENGINEER.

N. PROVIDE LAMINATED NAME PLATE ATTACHED TO EACH STARTER AND VFD IDENTIFYING THE SYSTEM IT SERVES.

O. STARTERS AND VFD'S SHALL BE PROVIDED WITH ENCLOSURES RATED NEMA 1 FOR INDOOR APPLICATIONS, NEMA 3R WITH ADDITIONAL GASKETING FOR WEATHERPROOF RAINTIGHT OUTDOOR ENCLOSURE OR INDOOR ENVIRONMENTS SUBJECT TO MOISTURE.

P. MOTORS SHALL BE HIGH EFFICIENCY, COMPLY WITH NEMA MG-1 STANDARD AND MEET THE 1992 EPA ENERGY EFFICIENCY ACT AND UTILITY COMPANY REBATE REQUIREMENTS.

Q. PROVIDE VARIABLE FREQUENCY DRIVES (VFD) AS MANUFACTURED BY GENERAL ELECTRIC, MAGNATEK, ABB, OR APPROVED EQUAL BY THE ENGINEER FOR CONTROL OF FANS AND/OR PUMPS AS SHOWN ON THE PLANS AND AS SPECIFIED HEREIN. VFD DISTORTION FACTOR SHALL NOT EXCEED 3% THD (VOLTAGE) AT POINT OF COMMON COUPLING, AS DEFINED BY IEEE 519.1992 AND IN NO CASE SHALL THE CURRENT THD EXCEED 10%. VFDS SHALL INCLUDE THE FOLLOWING:

1. PWM TECHNOLOGY INCORPORATING IGBT.
2. 40 CHARACTER FULL ENGLISH DIGITAL DISPLAY. CODES ARE NOT ACCEPTABLE.
3. DC LINE CHOKE.
4. MANUAL BYPASS AND MANUAL BYPASS CONTACTORS.
5. THREE SETS OF NORMALLY CLOSED OR NORMALLY OPEN CONTACTS.
6. CIRCUIT BREAKER DISCONNECT.
7. VFD DRIVE SERVICE SWITCH.
8. SPEED CONTROL DIAL.
9. THERMAL MOTOR OVERLOADS.
10. 3% AC LINE REACTOR PRE-WIRED AND INSTALLED WITHIN VFD ENCLOSURE.
11. FACTORY START-UP SERVICE INCLUDING COMPONENT TESTING, FIELD CHECK OF CONTROL CONNECTION, AND DOCUMENTS STATING THAT ALL WORK AND DRIVE FUNCTIONS ARE DEEMED ACCEPTABLE.
12. PROGRAMMING OF ALL DRIVE PARAMETERS PARTICULAR TO THIS INSTALLATION.
13. 2 YEAR SITE PARTS AND LABOR WARRANTY AFTER START-UP.

R. VARIABLE FREQUENCY DRIVE MOTORS SHALL COMPLY WITH NEMA MG-1 PART 31.40.4.2 STANDARD SUITABLE FOR VFD OPERATION. CONTRACTOR TO COORDINATE VFD AND MOTOR MANUFACTURERS.

S. ALL VFD DRIVES FOR ALL EQUIPMENT SHALL BE OF THE SAME MANUFACTURER. MECHANICAL CONTRACTOR SHALL COORDINATE VFD DRIVE MANUFACTURER WITH EACH EQUIPMENT VENDOR.

2.09 AUTOMATIC TEMPERATURE CONTROL

A. GENERAL

1. PROVIDE ALL CONTROL, POWER, AND INTERLOCK WIRING INCLUDING CONDUITS AND INSTALL PER THE NEW YORK CITY AND NATIONAL ELECTRIC CODE. SUBMIT TERMINAL TO TERMINAL WIRING DIAGRAM, SEQUENCE OF OPERATION AND CUTS OF ALL COMPONENTS FOR APPROVAL. PROVIDE ALL RELAYS, SWITCHES, DAMPERS AND ACTUATORS, PNEUMATIC EQUIPMENT, PILOT POSITIONERS THERMOSTATS, PANELS, LIMIT SAFETIES, TRANSFORMERS, TIME CLOCKS, CONTROL VALVES AND OTHER DEVICES TO ACCOMPLISH THE DESIRED SEQUENCE OF OPERATION.
2. FURNISH AND INSTALL AS HEREIN SPECIFIED, A COMPLETE AUTOMATIC TEMPERATURE CONTROL SYSTEM OF THE DDC TYPE WITH NATIVE BACNET DEVICES, MANUFACTURED BY SIEMENS INDUSTRY INC. 1000 DEERFIELD PARKWAY, BUFFALO GROVE, IL 60089, PHONE # (212) 258-4000. THE ATC CONTRACTOR SHALL BE AN INDEPENDENT CONTRACTOR NOT AFFILIATED WITH THE MECHANICAL CONTRACTOR.
3. ALL TEMPERATURE CONTROL SYSTEMS AND COMPONENTS ARE TO BE FULLY MODULATING TYPE, EXCEPT WHERE NOTED OTHERWISE.
4. IF NEW WORK IS TO CONNECT TO AN EXISTING SYSTEM, THE PROPOSED NEW SYSTEM TO BE INSTALLED SHALL BE FULLY COMPATIBLE WITH THE EXISTING SYSTEM. THE MANUFACTURER OF THE PROPOSED NEW SYSTEM SHALL PROVIDE ALL REQUIRED INTERFACES OR GATEWAYS TO ENSURE THAT THEIR SYSTEM IS FULLY COMPATIBLE.
5. WHEN CONNECTING TO AN EXISTING BMS WORKSTATION THE CONTRACTOR SHALL UPDATE THE EXISTING BMS WORKSTATION AND GRAPHICS WITH THE NEW SYSTEMS INSTALLED/MONITORED AS PART OF THIS PROJECT.
6. TRAINING
   A) THE BMS CONTRACTOR SHALL PROVIDE A MINIMUM OF (8) HOURS TRAINING FOR FACILITY STAFF ON-SITE. TRAINING CAN BE PERFORMED IN SEPARATE (4) HOUR INTERVALS AT THE DISCRETION OF THE OWNER.
   B) THE BMS CONTRACTOR SHALL PROVIDE HARDCOPY OF AS-BUILT DRAWINGS AND REVIEW ALL MAINTENANCE REQUIREMENTS AND PROCEDURES FOR ALL EQUIPMENT
7. ALL CONTROLS MUST BE THE PRODUCT OF ONE MANUFACTURER. ALL AUTOMATIC CONTROL VALVES AND DAMPER OPERATORS SHALL BE MANUFACTURED BY THE TEMPERATURE CONTROL MANUFACTURER.
8. THE MANUFACTURER OF THE AUTOMATIC CONTROL EQUIPMENT SHALL SUBMIT THE FOLLOWING FOR APPROVAL: A SCHEMATIC DIAGRAM OF EACH CONTROL SYSTEM WHICH SHALL INDICATE THE PROPER SEQUENCE OF OPERATION AND RANGE OF THE CONTROLS FOR ALL CYCLES, PROVIDE TERMINAL POINT TO TERMINAL POINT ELECTRICAL WIRING DIAGRAMS FOR APPROVAL, A COMPLETE DESCRIPTION OF THE AUTOMATIC OPERATION OF EACH SYSTEM WHERE THE DESCRIPTION INCLUDES THE DUTY OF EACH THERMOSTAT, VALVE, SWITCH, ETC., INCORPORATED IN THE CONTROL SYSTEM WITH A SCHEDULE AND ILLUSTRATION OF ALL CONTROL INSTRUMENTS AND EQUIPMENT INCLUDING CONTROL PANELS AND DEVICES FOR EACH SYSTEM.
9. INDIVIDUAL SMOKE DETECTORS SHALL BE INSTALLED (PROVIDED BY ELECTRICAL CONTRACTOR) IN THE RETURN DUCT OF ALL AIR HANDLING SYSTEMS SHARING A COMMON CEILING OR DUCT PLENUM AS REQUIRED BY CODE.
10. FOR AIR DISTRIBUTION SYSTEMS 2,000 CFM OR LARGER, INSTALL SMOKE DETECTORS (PROVIDED BY ELECTRICAL CONTRACTOR) IN MAIN SUPPLY DUCT (DOWNSTREAM OF AIR FILTERS AND AHEAD OF ANY BRANCH CONNECTIONS) AND MAIN RETURN DUCT (UPSTREAM OF ANY FILTERS AND BEFORE RETURN AIR IS DILUTED WITH OUTDOOR AIR).
11. PROVIDE SMOKE DAMPERS (N.C.) IN THE SUPPLY (AHEAD OF ANY BRANCH CONNECTIONS) AND RETURN DUCTS FOR SUPPLY AIR SYSTEMS 15,000 CFM OR LARGER AND SERVING MORE THAN THE FLOOR ON WHICH THE SYSTEM IS LOCATED. PROVIDE END SWITCHES IN SMOKE DAMPERS FOR CONNECTION TO FIRE ALARM SYSTEM AND HVAC CONTROLS.
12. ALL SMOKE DETECTORS SHALL BE TIED TO THE BUILDING FIRE ALARM SYSTEM. A SIGNAL FROM THE BUILDING FIRE ALARM SYSTEM SHALL AUTOMATICALLY SHUT DOWN SYSTEM FANS AND CLOSE ALL ASSOCIATED SMOKE DAMPERS AND COMBINATION FIRE/SMOKE DAMPERS. SIGNAL, INTERLOCK WIRING, POWER WIRING AND FINAL CONNECTIONS WILL BE PROVIDED BY ELECTRICAL CONTRACTOR.
13. FIRE/SMOKE DAMPERS SHALL REMAIN OPEN AT ALL TIMES DURING NORMAL OPERATION AND SHALL BE CONTROLLED BY THE AUTOMATIC TEMPERATURE CONTROL SYSTEM FOR THE HVAC SYSTEM CONTROL. FIRE/SMOKE DAMPERS SHALL BE UNDER THE FULL CONTROL OF THE FIRE ALARM SYSTEM. CONTROL CONTRACTOR SHALL COORDINATE WITH THE ELECTRIC CONTRACTOR FOR ALL CONTROL WIRING, DEVICES, AND INTERLOCK WIRING TO OPEN, CLOSE AND PROVE FIRE SMOKE DAMPER POSITIONS.
14. ALL AUTOMATIC TEMPERATURE CONTROL SYSTEM SHALL COMPLY WITH THE NEW YORK CITY ENERGY CONSERVATION CODE REQUIREMENTS. ALL HVAC SYSTEM CONTROLS SHALL COMPLY WITH THE FOLLOWING REQUIREMENTS:
   A) HEATING AND COOLING TO EACH ZONE SHALL BE CONTROLLED BY A THERMOSTAT. A MINIMUM OF ONE HUMIDITY CONTROL DEVICE SHALL BE INSTALLED PER HUMIDIFICATION OR DEHUMIDIFICATION SYSTEM.
   B) ALL ZONE THERMOSTAT SHALL OPERATE WITH A MINIMUM OF 5°F DEADBAND BETWEEN HEATING AND COOLING WITH SETPOINT OVERLAP CAPABILITY.
   C) ALL ZONE THERMOSTATS SHALL BE OPERATED VIA THERMOSTATIC SETBACKS CONTROLS OPERATED VIA AN AUTOMATIC TIME CLOCK OR PROGRAMMABLE CONTROL SYSTEM.
   D) ALL CONTROLS SHALL HAVE THE ABILITY TO SETBACK TEMPERATURE DOWN TO 55°F OR UP TO 85°F.
   E) CONTROLS SHALL BE CAPABLE OF AUTOMATICALLY STARTING AND STOPPING THE SYSTEM FOR SEVEN DIFFERENT DAILY SCHEDULES PER WEEK, CAPABLE OF HAVING SETTING SAVED IN MEMORY FOR 10 HOURS DURING A LOSS OF POWER AND A MANUAL SYSTEM “ON” OVERRIDE FOR UP TO TWO (2) HOURS OR AN OCCUPANCY SENSOR.
   F) CONTROL SYSTEM SHALL AUTOMATICALLY RESET SUPPLY AIR TEMPERATURE IN RESPONSE TO BUILDING LOAD OR OUTSIDE AIR TEMPERATURE.
   G) THE CONTRACTOR SHALL PROVIDE TESTING TO ENSURE PER OPERATION, CALIBRATION AND ADJUSTMENT OF CONTROLS.
   H) OFF-HOURS CONTROLS WITH SETBACK AND/OR SHUTDOWN CAPABILITIES. FOR SPECIFIC OCCUPANCIES AND CONDITIONS, EACH SPACE-CONDITIONING SYSTEM MUST BE PROVIDED WITH CONTROLS THAT CAN AUTOMATICALLY SHUT-OFF THE EQUIPMENT DURING UNOCCUPIED HOURS. THE CONTROL DEVICE SHALL BE AN AUTOMATIC WITH SWITCH DEVICE. THIS CAN BE ACCOMPLISHED WITH A 7-DAY PROGRAMMABLE THERMOSTAT WITH BACKUP CAPABILITIES THAT STORES THE DEVICE'S SCHEDULE FOR AT LEAST 7 DAYS AND THE TIME AND DATE FOR AT LEAST 72 HOURS IF POWER IS LOST.

B. ELECTRIC WIRING:

1. ALL ELECTRIC WORK (EXCEPT FOR MOTOR FEEDERS, WIRING BETWEEN MOTORS, MOTOR CONTROLLERS, FEEDER PANELS, FUSES, CIRCUIT BREAKERS AND BUS BARS) REQUIRED FOR THE AUTOMATIC TEMPERATURE CONTROL SYSTEM SHALL BE PROVIDED BY THIS CONTRACTOR. WORK SHALL INCLUDE BUT NOT BE LIMITED TO TIME SWITCHES, DAMPER MOTORS, DAMPER SWITCHES, ELECTRIC THERMOSTAT, ELECTRIC RELAYS, E/P SWITCHES, INTERLOCKING WIRING, WIRE, CONDUIT, ETC.
1. ALL CONTROL POWER, WIRING AND TRANSFORMERS FOR DAMPERS, ACUATORS, VAV BOXES, CONTROL PANLES, ETC. TO BE PROVIDED BY THE CONTROLS CONTRACTOR FROM A SOURCE DESIGNATED BY THE ELECTRICAL CONTRACTOR. CIRCUITS FOR CONTROL DEVICES HAVE BEEN DESIGNATED IN THE ELECTRICAL PANEL SCHEDULES.
2. THE CONTROL MANUFACTURER SHALL INCLUDE WIRING DIAGRAMS IN HIS SHOP DRAWINGS SUBMITTALS FULLY COORDINATED WITH THE ELECTRICAL CONTRACTORS WORK. IT SHALL BE THE AUTOMATIC TEMPERATURE CONTROL CONTRACTORS RESPONSIBILITY TO PROVIDE ALL WIRING AND CONDUIT AS REQUIRED TO ACHIEVE THE FUNCTION CALLED FOR IN THESE SPECIFICATIONS, CONFORMING WITH LOCAL CODES FOR MATERIAL AND INSTALLATION. THE ELECTRICAL SPECIFICATION FOR THE PROJECT ELECTRICAL WORK IS TO BE FOLLOWED.

G. CONTROL PANELS SHALL BE NEMA 1 FOR INDOOR APPLICATIONS, NEMA 3R WITH ADDITIONAL GASKETING FOR WEATHERPROOF RAINTIGHT OUTDOOR ENCLOSURE OR INDOOR ENVIRONMENTS SUBJECT TO MOISTURE. THEY SHALL BE PROVIDED WITH WELDED ANGLE BRACKETS AND A BAKED PRIME COAT ENAMEL FINISH. THE PANEL DOORS SHALL BE HINGED LOCKING TYPE. CONTROL PANELS SHALL CONTAIN ALL CENTRAL CONTROL DEVICES, SUCH AS CONTROLLERS, RELAYS, SWITCHES, PILOT LIGHTS, TERMINAL BLOCKS, AND ALL OTHER ACCESSORIES AS REQUIRED FOR A WORKABLE ENVIRONMENTAL CONTROL SYSTEM. ALL COMPONENTS WITHIN THE CONTROL PANELS SHALL BE PRE-WIRED TO NUMBERED TERMINAL TRIPS, READY FOR FIELD CONNECTION FOR FIELD MOUNTED CONTROL COMPONENTS. PROVIDE ENGRAVED NAMEPLATES TO LABEL THE CONTROLLED EQUIPMENT. PROVIDE A PLASTIC LAMINATED CONTROL SCHEMATIC DRAWING HUNG NEXT TO EACH CONTROL PANEL.

H. THE SYSTEM INSTALLATIONS SHALL BE SUPERVISED BY THE AUTOMATIC CONTROL MANUFACTURER, WHO SHALL COORDINATE WITH AND INSTRUCT PIPING OR SHEET METAL TRADES AS TO TEES OR TAPPINGS TO BE INSTALLED IN PIPING OR EQUIPMENT AND OPENINGS THAT ARE REQUIRED IN SHEET METAL FOR THE SETTING AND INSTALLATIONS OF CONTROL DEVICES THEREIN BY THESE TRADES.

I. ALL ROOM THERMOSTATS/SENSORS AND SWITCH LOCATIONS SHALL BE SUBMITTED FOR REVIEW BY THE ARCHITECT AND ENGINEER PRIOR TO INSTALLATION WHETHER THE DEVICES ARE SHOWN ON PLANS OR NOT.

J. ALL ROOM THERMOSTATS/SENSORS SHALL MATCH THE BUILDING STANDARD.

K. AUTOMATIC VALVES:

1. ALL AUTOMATIC CONTROL AND ISOLATION VALVES SHALL BE OF THE ELECTRONIC TYPE, UNLESS OTHERWISE SPECIFIED, QUIET IN OPERATION, AND SHALL BE ARRANGED TO SPRING RETURN FAIL SAFE, IN A NORMALLY OPEN OR NORMALLY CLOSED POSITION. CONTROL VALVES SHALL BE FULLY PROPORTIONING AND ISOLATION VALVES SHALL BE 2-POSITION. VALVES TO HAVE ADJUSTABLE OPERATING RANGES AND STARTING POINTS TO PROVIDE FLEXIBILITY OF ADJUSTMENT IN SEQUENCING AND THROTTLING. MODULATING VALVES SHALL BE PROVIDED WITH PILOT POSITIONERS.
2. VALVES SHALL BE SIZED BY THE TEMPERATURE CONTROL MANUFACTURER AND GUARANTEED TO MEET THE HEATING OR COOLING REQUIREMENTS AS SPECIFIED. CONTROL VALVES SHALL BE SUITABLE FOR PRESSURE CONDITIONS AND CLOSE AGAINST 110% OF PUMP DIFFERENTIAL PRESSURE.
3. ALL VALVE BODIES SHALL HAVE THE SAME PRESSURE CHARACTERISTICS AS THE PIPE IN WHICH IT IS INSTALLED.
4. VALVES 2 INCHES AND SMALLER UNLESS OTHERWISE SPECIFIED SHALL HAVE BRONZE BODIES WITH SCREWED CONNECTIONS. VALVES SHALL BE FISHER TYPE ED, WARREN TYPE 20/70, K&M SERIES GCG, OR AS APPROVED.
5. VALVES BETWEEN 2-1/2" AND 4 INCH UNLESS OTHERWISE SPECIFIED, SHALL HAVE CAST IRON OR CARBON STEEL BODIES WITH FLANGED CONNECTIONS IN ACCORDANCE WITH THE PIPING SPECIFICATIONS. VALVES SHALL BE FISHER STYLE ED, WARREN TYPE 20/70 OR 1800 SERIES GCG, K&M SERIES GCG OR AS APPROVED.
6. AUTOMATIC CONTROL VALVES EXPOSED TO THE ELEMENTS SHALL HAVE ELECTRONIC ACTUATORS WITH ALL REQUIRED ACCESSORIES.

I. AUTOMATIC DAMPERS:

1. PROVIDE CONTROLS FOR ALL THE AUTOMATIC DAMPERS, AS SPECIFIED IN THE DUCTWORK SECTION, AND SHOWN ON THE DRAWINGS.
2. CONTROL MOTORS OR ACTUATORS SHALL BE OF THE ELECTRONIC TYPE, UNLESS OTHERWISE NOTED, OF APPROPRIATE SIZE AND QUANTITIES TO PROVIDE TWO-POSITION OR PROPORTIONING CONTROL ACTION AS SPECIFIED. PROPORTIONING TYPE SHALL BE EQUIPPED WITH PILOT TYPE POSITIONERS. PILOT POSITIONERS SHALL BE SELECTED FOR VARIED SPRING RANGES AND ADJUSTABLE WITHOUT DISMANTLING POSITIONER AND CONTROL MOTOR.
3. AUTOMATIC DAMPERS EXPOSED TO THE ELEMENTS SHALL HAVE ELECTRIC ACTUATORS WITH ALL REQUIRED ACCESSORIES.

I. SEQUENCES OF OPERATION - FURNISH AND MOUNT ALL DEVICES AS REQUIRED TO PERFORM THE FOLLOWING SEQUENCES OF OPERATION:

1. SERIES FAN POWERED VARIABLE AIR VOLUME (VAV) BOXES:
   a) ELECTRICALLY CONTROLLED FPVAV BOXES WITH FAIL IN PLACE DAMPERS SHALL BE CONTROLLED FROM A UNIT MOUNTED DDC CONTROLLER.
   b) A DDC SOFTWARE PROGRAM SHALL START THE FPVAV BOXES INDIVIDUALLY WHEN THE RESPECTIVE BASE BUILDING AC SYSTEM IS IN OPERATION.
   c) PROVIDE ROOM THERMOSTATS (SENSORS), ONE PER VAV BOX, LOCATED AS INDICATED ON THE DRAWINGS TO MAINTAIN A 75°F SPACE TEMPERATURE (ADJUSTABLE).
      (1) WHEN THE SPACE TEMPERATURE RISES, THE CONTROLLER SHALL MODULATE THE PRIMARY AIR VALVE TO ITS MAXIMUM POSITION AS REQUIRED TO MAINTAIN THE SPACE TEMPERATURE SETPOINT.
      (2) AS THE SPACE TEMPERATURE DECREASES, THE TERMINAL UNIT DAMPER SHALL MODULATE TO ITS MINIMUM POSITION OF 30%, UNLESS OTHERWISE NOTED.
      (3) WHEN THE SPACE TEMPERATURE CONTINUES TO DECREASE, THE OUTPUT OF THE CONTROLLER SHALL MODULATE THE HEATING COIL CONTROL VALVE TO MAINTAIN SPACE TEMPERATURE. THIS CONTRACTOR SHALL PROVIDE THE CONTROL VALVE, ACTUATOR AND CONTROLS.
      (4) THE VALVE SHALL FAIL TO THE FULL HEAT POSITION.
   d) AS THE ROOM TEMPERATURE VARIES BETWEEN +/-1°F OF THE THERMOSTAT SETPOINT, THE CONTROLLER RESETS AIR FLOW BETWEEN MAXIMUM AND MINIMUM.
   e) DURING WARM-UP, THE FPVAV SHALL REMAIN UNDER CONTROL.
   f) WHENEVER THE FPVAV IS OFF, THE TERMINAL UNIT DAMPER SHALL MODULATE TO ITS MINIMUM FLOW POSITION.
   g) WHEN THE FPVAV IS OFF, THE SET POINT OF THE CONTROLLER SHALL BE LOWERED TO 55 DEGREES F ADJUSTABLE. IF THE TEMPERATURE FALLS BELOW SET POINT, THE CONTROLLER SHALL START THE FAN AND MODULATE THE REHEAT COIL TO MAINTAIN SPACE TEMPERATURE SET POINT.
   h) MECHANICAL CONTRACTOR TO COORDINATE THE PROVISION OF ALL POWER, CONTROL WIRING, TRANSFORMERS, DISCONNECT SWITCHES, ETC. WITH A.T.C. CONTRACTOR AS REQUIRED.
   i) THERMOSTATS/SENSORS:
      (1) THERMOSTATS/SENSORS SHALL BE PROVIDED FOR EACH VAV BOX.
      (2) ALL ROOM THERMOSTATS/SENSORS SHALL MATCH THE BUILDING STANDARD.
   j) DURING AUTOMATIC FIRE/SMOKE PURGE, THE VAV BOX DAMPER SHALL OPEN.
   k) FOR ALL BOXES SERVING AN ENCLOSED ROOM GREATER THAN 500 SQ FT, SENSORS SHALL MONITOR THE CO2 LEVEL IN THE SPACE AND REPORT TO THE BMS SYSTEM. BMS SHALL BE ALARMED IF CO2 LEVEL RISES ABOVE ITS ADJUSTABLE SETPOINT.
   l) VAV BOXES CONTROLLED BY MULTIPLE ROOM SENSORS SHALL BE CAPABLE OF BEING CONTROLLED BASED ON EITHER THE WORST-CASE TEMPERATURE OR AN AVERAGE OF THE MULTIPLE SENSORS (OPERATOR SELECTABLE VIA THE BMS) AS REQUIRED.
   m) CONTROLLERS AND ACTUATORS SHALL BE SHIPPED TO THE TERMINAL BOX MANUFACTURER FOR FACTORY MOUNTING. STEEL CONTROL ENCLOSURES TO BE PROVIDED BY BOX MANUFACTURER.
   n) EACH BOX SHALL BE CONNECTED TO THE LANDLORD BMS AND BE CAPABLE OF PERFORMING THE FOLLOWING MINIMUM REMOTE FUNCTIONS:
      (1) COOLING SETPOINT ADJUSTMENT.
      (2) INDEX BOX TO MINIMUM OR MAXIMUM FLOW QUANTITY.
      (3) MONITOR ZONE TEMPERATURE, SUPPLY AIR TEMPERATURE AND PRIMARY AIRFLOW THROUGH VAV BOX.
      (4) MONITOR FAN STATUS
      (5) FAN COMMAND ON/OFF
2. EXHAUST/TRANSFER FANS:
   a) FANS SERVING PANTRIES, TOILET ROOMS, ETC. SHALL OPERATE CONTINUOUSLY DURING OCCUPIED PERIODS. TIME SCHEDULE SHALL MATCH THE BUILDING HOURS OF OPERATION AND BE ADJUSTABLE.
   b) FANS SERVING IDF ROOMS, AV CLOSETS ETC. SHALL BE STARTED AND STOPPED VIA A WALL MOUNTED THERMOSTAT.
3. CONDENSER WATER PUMPS
   a) CIRCULATING PUMP OPERATION SHALL BE INTERLOCKED WITH THE AC UNITS. UPON ACTIVATION OF ANY UNIT, THE LEAD CIRCULATING PUMP SHALL START AND RUN CONTINUOUSLY. THE PUMPS SHALL BE PROVIDED WITH AN AUTOMATIC RESTART AFTER A POWER OUTAGE.
   b) PROVIDE FOR LEAD/LAG SEQUENCING IN ORDER TO HAVE ONE PUMP RUNNING AND THE OTHER AS STAND-BY. PUMPS SHALL BE CYCLED ON A 150 HOUR OPERATING SCHEDULE (ADJUSTABLE) FOR EVEN WEAR.
   c) A FLOW SENSOR (ON DISCHARGE OF EACH PUMP) SHALL PROVE FLOW BEFORE STARTING ANY AIR CONDITIONING UNITS. A LOSS OF FLOW AS INDICATED BY THE FLOW SENSOR ON THE RUNNING PUMP SHALL SWITCH OPERATION TO THE STAND-BY PUMP AND ANNUNCIATE AN AUDIO/VISUAL ALARM LOCALLY.
   d) PROVIDE A LEAK DETECTOR AT THE AUXILIARY DRAIN PAN OF EACH PUMP. LEAK DETECTION SHALL ANNUNCIATE AN AUDIO/VISUAL ALARM LOCALLY AND THROUGH THE LANDLORD BMS. IF MOISTURE IS DETECTED IN THE PAN OF THE OPERATING PUMP, THE PUMP SHALL STOP, AND THE STAND-BY PUMP SHALL START.
   e) SYSTEM DIFFERENTIAL PRESSURE WILL BE MAINTAINED BY VARIABLE SPEED DRIVES PROVIDED FOR EACH PUMP. THE PUMP SPEED SHALL MODULATE AS REQUIRED TO MAINTAIN A SYSTEM DIFFERENTIAL PRESSURE SETPOINT (ADJUSTABLE) AS SENSED BY A PIPE MOUNTED DIFFERENTIAL PRESSURE SENSOR. DIFFERENTIAL PRESSURE SENSOR TO BE MOUNTED 3/4 DOWNSTREAM IN THE LONGEST PIPING DISTRIBUTION RUN.
   f) PROVIDE CONDENSER WATER PUMP SYSTEM WITH A 24 HOUR, 7 DAY TIMER TO INSURE PUMP OPERATION NOT LESS THAN 1 HOUR EVERY 4 HOURS.
4. SUPPLEMENTAL WATER COOLED AIR CONDITIONING UNITS:
   a) AIR CONDITIONING UNITS SERVING IDF OR FIBER CLOSETS SHALL OPERATE 24 HOURS A DAY, SEVEN DAYS PER WEEK IN TIMED SCHEDULE.
   b) AIR CONDITIONING UNITS SERVING GENERAL OFFICE SPACE SHALL BE STARTED/STOPPED NORMALLY BY AN ADJUSTABLE PRESET TIMED SCHEDULE AS DETERMINED BY OWNER (E.G. 8:00 AM TO 6:00 PM).
   c) UNITS SHALL BE AUTOMATICALLY STARTED AND STOPPED VIA AN INDIVIDUAL DIGITALLY PROGRAMMABLE COOLING THERMOSTAT WITH SEVEN-DAY TIME CLOCK CONTROL AND LOCKING COVER LOCATED WITHIN THE SERVED SPACE. THE TIME SCHEDULE TO BE DETERMINED BY OWNER. THERMOSTATS TO INCLUDE TIMED OVERRIDE CAPABILITY TO OVERRIDE THE CLOCK SETTING IN THREE (3) HOUR INCREMENTS (ADJUSTABLE). THERMOSTATS TO INCLUDE MULTIPLE STAGES AS REQUIRED BY EQUIPMENT CONTROLLED. THERMOSTAT SHALL BE SIMILAR TO HONEYWELL TB8220 PREMIUM THERMOSTAT.
   d) THERMOSTAT SHALL HAVE LOCAL SETPOINT AND ACTUAL SPACE TEMPERATURE READOUTS AND SHALL HAVE A SETPOINT OF 75°F ADJ. THE AC UNIT CONTROLS SHALL CYCLE AND MODULATE THE REFRIGERATION COMPONENTS TO MAINTAIN THE SPACE TEMPERATURE SETPOINT.
   e) THE THERMOSTAT SHALL ALLOW THE UNITS FAN TO RUN CONTINUOUSLY WHILE IN OPERATION.
   f) UNIT SHALL MODULATE WATER REGULATING VALVE AS REQUIRED TO MAINTAIN REFRIGERANT PRESSURES.
   g) PROVIDE NORMALLY CLOSED MOTORIZED ISOLATION VALVES ON THE CONDENSER WATER SUPPLY AND RETURN PIPING LINES.
   h) MOTORIZED ISOLATION VALVES SHALL OPEN UPON AC UNIT ACTIVATION AND CLOSE WHEN THE UNIT IS NOT OPERATING. VALVES SHALL OPEN DAILY FOR 4 HOURS (ADJUSTABLE) TO FLOW WATER THROUGH THE CONDENSER COIL REGARDLESS OF UNIT OPERATION.
   i) THE RESPECTIVE CONDENSATE PUMP FOR EACH AIR CONDITIONING UNIT SHALL BE ENERGIZED AT ALL TIMES REGARDLESS OF THE AIR CONDITIONING UNIT'S OPERATION. WHEN THE CONDENSATE PUMPS SAFETY FLOAT SWITCH IS ENGAGED, AN ALARM SIGNAL SHALL SOUND AT THE LANDLORD BMS. IF THE ALARM IS NOT RESET WITHIN 5 MINUTES (ADJUSTABLE), THE AIR CONDITIONING UNIT SHALL BE DE-ENERGIZED AND THE ISOLATION VALVES CLOSED.
   j) LEAK DETECTION SHALL BE PROVIDED IN THE AUXILIARY DRAIN PAN FOR EACH UNIT. WHEN THE LOCAL LEAK DETECTOR SENSES MOISTURE IN THE AUXILIARY DRAIN PAN, AN ALARM SIGNAL SHALL SOUND AT THE LANDLORD BMS. IF THE ALARM IS NOT RESET WITHIN 5 MINUTES (ADJUSTABLE), THE AIR CONDITIONING UNIT SHALL BE DE-ENERGIZED AND THE ISOLATION VALVES CLOSED.
   k) PROVIDE A FLOW SWITCH FOR EACH AIR CONDITIONING UNIT. THE FLOW SWITCH SHALL SENSE THE MINIMUM CONDENSER WATER FLOW NECESSARY FOR OPERATING THE UNIT. UPON REACHING THE MINIMUM FLOW SETPOINT, THE FLOW SWITCH SENSOR SHALL ENGAGE A CONTACT AND ALLOW THE UNIT TO OPERATE. IF THE FLOW RATE IS BELOW THE MINIMUM SETPOINT, THE CONTACT SHALL OPEN AND PREVENT THE UNIT FROM RUNNING:
   l) UPON A CALL TO ACTIVATE ANY AC UNIT, THE RESPECTIVE CONDENSER WATER PUMP SHALL START AND RUN CONTINUOUSLY. REFER TO THE CONDENSER WATER PUMP SEQUENCE OF OPERATION FOR ADDITIONAL REQUIREMENTS.
5. LEAK DETECTION
   a) A LEAK DETECTOR SHALL BE PROVIDED IN ALL AUXILARY DRAIN PANS. DRAIN PANS SHALL BE PROVIDED FOR EACH AC UNIT, DOMESTIC WATER HEATER, PUMP, ETC. INCLUDED IN THIS PROJECT. REFER TO MECHANICAL AND PLUMBING DRAWINGS FOR QUANTITIES AND LOCATIONS.
   b) A LEAK DETECTOR SHALL BE PROVIDED ON THE FLOOR WITHIN A FIVE FOOT RADIUS OF ALL PANTRY EQUIPMENT AND PLUMBING FIXTURES THAT HAVE A DOMESTIC WATER CONNECTION. THIS INCLUDES SINKS, DISHWASHERS, COFFEE MAKERS, ICE MAKERS, BEVERAGE DISPENSERS, ETC. REFER TO ARCHITECTURAL DRAWINGS FOR QUANTITIES AND LOCATIONS.
   c) A LEAK DETECTOR SHALL BE PROVIDED UNDER THE LAVATORY IN EACH TOILET AND SHOWER ROOM WITH THE EXCEPTION OF THE BASE BUILDING CORE TOILET ROOMS. REFER TO ARCHITECTURAL DRAWINGS FOR QUANTITIES AND LOCATIONS.
   d) UPON SENSING A LEAK, THE LEAK DETECTOR SHALL BE HARDWIRED TO SHUT DOWN THE ASSOCIATED EQUIPMENT, CLOSE THE ISOLATION VALVES TO STOP WATER FLOW, AND SEND AN ALARM TO THE LANDLORD BMS AND TENANT ALARM SYSTEM.
   e) ALL AC UNITS SHALL BE PROVIDED WITH A CONDENSATE PUMP WITH OVERFLOW ALARM CONTACTS WIRED TO ALARM THE LANDLORD BMS AND TENANT ALARM SYSTEM
6. TENANT ALARM SYSTEM
   a) TENANT SHALL BE NOTIFIED BY AN EMAIL MESSAGE OF ALL LEAK ALARMS.
   b) ALARM MESSAGE SHALL INDICATE ADDRESS, FLOOR #, SYSTEM SERVED, AND ROOM # WHERE LEAK HAS OCCURRED.
   c) A TOTAL OF 5 EMAIL ADDRESSES SHALL BE NOTIFIED OF ALARMS.

PART 3- EXECUTION

3.01 A. PROVIDE AND INSTALL ALL EQUIPMENT AND ACCESSORIES OF THE SIZES AND CAPACITIES AS SCHEDULED AND AS INDICATED ON THE DRAWINGS AND IN ACCORDANCE WITH APPROVED SHOP DRAWINGS AND MANUFACTURERS RECOMMENDATIONS. PROVIDE ALL MOTOR STARTERS AS REQUIRED; MOTOR STARTERS WILL BE INSTALLED BY THIS CONTRACTOR AND WIRED BY ELECTRICAL TRADE.

A. CONTRACTOR IS RESPONSIBLE FOR MAINTAINING ALL REQUIRED CLEARANCES FOR SERVICING AND MAINTENANCE. COORDINATE REQUIREMENTS WITH ALL TRADES.

B. IDENTIFICATION OF EQUIPMENT AND CONTROLS:

1. ALL EQUIPMENT SHALL BE STENCILED OR LABELED WITH LAMACOID PLATES SCREWED THEREON WHICH SHALL INDICATE SYSTEMS SERVICE.
2. MOTOR STARTERS SHALL BE PROVIDED WITH LAMACOID PLATES WHICH INDICATE SYSTEM SERVED.
3. CONTRACTOR TO SUBMIT LIST OF EQUIPMENT TO RECEIVE LABELS AND THE COORDINATED DESIGNATIONS, SIZE OF LABEL LETTERING, PLATE SIZE AND COLOR FOR REVIEW PRIOR TO INSTALLATION.

C. FOR ALL FLOOR MOUNTED EQUIPMENT PROVIDE A 4" HIGH CONCRETE HOUSE-KEEPING PAD; WHERE FLOOR STANDS ARE INDICATED PROVIDE FLOOR STAND OF STRUCTURAL STEEL OR STEEL PIPES AND FITTINGS AND BOLT TO PAD; FOR ROOF MOUNTED EQUIPMENT PROVIDE SUPPORTS WITH APPROVED ANCHORS DIRECTLY FROM BUILDING STEEL STRUCTURE. PROVIDE SUPPLEMENTARY STEEL AS REQUIRED TO ADEQUATELY SUPPORT THE LOAD.

3.02 CHEMICAL CLEANING AND PRETREATMENT

A. CLEANING OF PIPING SHALL BE PERFORMED IN THE PRESENCE OF A BUILDING REPRESENTATIVE.

B. PROVIDE ALL DISPERSANTS, SCALE INHIBITORS AND CORROSION INHIBITORS AS REQUIRED FOR CLEANING AND TREATING ALL PIPING SYSTEMS. CHROMATES SHALL NOT BE USED.

C. ALL CHEMICALS TO BE USED FOR PIPE CLEANING SHALL BE APPROVED BY THE BASE BUILDING CHEMICAL TREATMENT COMPANY.

D. FLUSH PIPING SYSTEMS WITH THE APPROVED CLEANING CHEMICAL TO REMOVE PIPE DOPE, SLUSHING COMPOUNDS, CUTTING OILS AND OTHER LOOSE EXTRANEOUS MATERIALS. SEAL ENDS AFTER CLEANING.

E. THE CONTRACTOR SHALL:

1. SATISFY EACH CHEMICAL HAS THE PROPER FEED RATES FOR CLEANING AND PRETREATMENT OF EACH SYSTEM AND RECORD.
2. CHECK THAT THE CLEANING SOLUTION IS ACTUALLY IN EACH SYSTEM.
3. SATISFY WHEN TO FLUSH THE SYSTEM.
4. CHECK EACH SYSTEM FOLLOWING FLUSHING TO ENSURE CLEANING CHEMICALS HAVE BEEN REMOVED FROM EACH SYSTEM AND TEST TO ENSURE PH OF NEW SYSTEM IS WITHIN 0.5 OF FRESH INCOMING WATER.

F. BLOCK MODULATING VALVES, ZONE VALVES AND OTHER SYSTEM RESTRICTIONS. PROVIDE BY PASS PIPING AND VALVING TO ISOLATE NEW AND EXISTING TO BE RE-USED EQUIPMENT SUCH AS CHILLERS, COILS, HEAT EXCHANGERS, ETC. FROM THE CLEANING PROCESS.

G. PROVIDE PORTABLE PUMPS TO CIRCULATE WATER FOR CLEANING PURPOSES AT RESPECTIVE FLOWS FOR FOUR (4) HOURS. REMOVE AND CLEAN STRAINERS. BLOW OFF LOW POINTS WITH STEAM AFTER CLEANING AND BEFORE TRAPS ARE INSTALLED. DRAIN ENTIRE SYSTEM.

H. CHEMICAL USED FOR CLEANING OF SYSTEMS SHALL COMPLY WITH THE RECOMMENDATIONS OF THE MANUFACTURERS OF THE MAJOR COMPONENTS IN THE SYSTEM AND SHALL BE APPROVED FOR USE.

I. UPON INITIAL FILL (FOLLOWING SYSTEM FLUSHING) THE APPROVED CHEMICALS WHICH PROVIDE A PROTECTIVE COATING TO PREVENT OXIDATION OF THE CLEANED SYSTEM SHALL BE ADDED.

3.03 WATER TREATMENT

C. ALL WATER TREATMENT REQUIREMENTS TO BE PROVIDED BY EXISTING BASE BUILDING WATER SYSTEM.

3.04 EQUIPMENT START-UP AND TESTING

A. UPON COMPLETION OF THE INSTALLATION, THIS CONTRACTOR SHALL ENSURE THAT ALL EQUIPMENT AND SYSTEMS ARE TESTED AND BALANCED UNDER FIELD OPERATING CONDITIONS TO DEMONSTRATE ITS COMPLIANCE WITH SPECIFICATION REQUIREMENTS.

B. SHOULD ANY PART OF THE EQUIPMENT OR SYSTEM FAIL TO MEET THE CONTRACT REQUIREMENTS, THIS CONTRACTOR SHALL ADJUST, REPAIR OR REPLACE ALL DEFECTIVE OR INOPERATIVE PARTS AND AGAIN CONDUCT THE COMPLETE START-UP TEST.

C. SUBMIT SYSTEM START UP SHEETS AND TEST RESULTS TO THE OWNER AND ENGINEER.

3.05 AIR AND WATER BALANCING

A. AIR AND WATER SYSTEM BALANCING SHALL BE PERFORMED BY ADVANCED TESTING & BALANCING (CONTACT KEVIN WILTON 718-486-6764) OR AN APPROVED INDEPENDENT CERTIFIED TESTING AND BALANCING FIRM. THE TESTING AND BALANCING FIRM SHALL BE AABC, NEBB, TABB CERTIFIED OR DIRECTLY SUPERVISED BY A STAFFED LICENCED PROFESSIONAL ENGINEER WITH A MINIMUM OF FIVE YEARS EXPERIENCE. AIR AND WATER SYSTEM BALANCING SHALL BE PERFORMED IN THE PRESENCE OF A BUILDING REPRESENTATIVE.

B. MAKE ALL REQUIRED ADJUSTMENTS OF ALL NEW AND EXISTING AIR AND WATER SYSTEM DEVICES UNTIL ALL SPECIFIED PERFORMANCES ARE MET. PROVIDE NECESSARY PIPING AND CONNECTIONS FOR BALANCING ALL WATER SYSTEMS. PROVIDE VOLUME DAMPERS AS REQUIRED FOR FINAL BALANCING OF AIR SYSTEMS. PROVIDE A CLEAN SET OF AIR FILTERS AT ALL AIR CONDITIONING UNITS AND CLEAN ALL STRAINERS PRIOR TO ANY BALANCING.

C. SUBMIT THREE (3) AIR AND WATER BALANCING REPORTS FOR REVIEW CONSISTING OF DESIGN AND ACTUAL READINGS OF ALL EQUIPMENT/DEVICES, LOCATION PLANS OF ALL EQUIPMENT/DEVICES BALANCED, BALANCING EQUIPMENT USED AND METHODS OF BALANCING.

D. ALL REPORTS SHALL INDICATE PRELIMINARY READINGS PRIOR TO BALANCING AND FINAL READINGS AFTER BALANCING HAS BEEN COMPLETED. IF IT IS DETERMINED THAT DRIVE CHANGES ARE REQUIRED, CONTRACTOR SHALL PROVIDE ALL NECESSARY NEW COMPONENTS.

E. CONTRACTOR SHALL INCLUDE IN THEIR BID TWO (2) JOB SITE COMFORT BALANCES UPON ACCEPTANCE OF THE FINAL BALANCING REPORT.

F. CONTRACTOR SHALL SUBMIT WATER BALANCE DATA SHEETS AND REPORTS WHICH TABULATE TEST DATA FOR FINAL ADJUSTED SYSTEM CONDITIONS WITHIN 2% OF DESIGN QUANTITIES FOR SYSTEM COMPONENTS INDICATING GPM AND PRESSURE DROP AT PIPE RISERS AND MAINS; PERFORMANCE CHARACTERISTICS FOR ALL PUMPS INDICATING RPM, GPM, TDH, AMPS, SUCTION AND DISCHARGE HEAD PRESSURE, BHP AND HP AT DESIGN AND NO FLOW CONDITIONS; PRESSURE DROP ACROSS COILS, EQUIPMENT, EACH RISER AND MAIN. MARK BALANCING VALVE TAG OF BALANCED POSITION.

G. CONTRACTOR SHALL SUBMIT AIR BALANCE DATA SHEETS AND REPORTS WHICH TABULATE TEST DATA FROM FINAL ADJUSTED SYSTEM CONDITIONS WITHIN 10% OF DESIGN QUANTITIES FOR SYSTEM COMPONENTS AIR OUTLETS, RETURNS AND TERMINAL UNITS INDICATING CFM AND PRESSURE DROP AT DUCT RISERS AND MAINS; PERFORMANCE CHARACTERISTICS FOR ALL FANS AND AIR CONDITIONING EQUIPMENT INDICATING RPM, CFM, PRESSURE DROP ACROSS EACH COMPONENT (FILTERS, COILS, DAMPERS, ETC), AMPS, SUCTION AND DISCHARGE STATIC PRESSURE, OUTSIDE AIR CFM, BHP AND HP AT DESIGN CONDITIONS; AIR OUTLET DISCHARGE TEMPERATURE AND CFM; TERMINAL BOX INLET SP, MINIMUM AND MAXIMUM AIR SETTINGS.

H. BALANCING CONTRACTOR SHALL BALANCE THE SYSTEM AND ALL ITS COMPONENTS IN SUCH A WAY AS TO ENSURE THE COMPONENTS ARE TESTED AS PER ALL THE ANTICIPATED OPERATING CONDITIONS.

I. BALANCING OF INDIVIDUAL COMPONENTS WITHOUT THE ENTIRE SYSTEM OPERATING AS INTENDED PER THE SEQUENCE OF OPERATIONS SHALL BE DEEMED UNACCEPTABLE AND ANY ISSUES WITH ACHIEVING AIR OR WATER FLOWS RESULTING FROM BALANCING IN THIS MANNER SHALL BE CORRECTED BY THE INSTALLING CONTRACTOR AS NECESSARY AT NO COST TO THE CLIENT.

J. CONTRACTOR TO PROVIDE TRAVERSE READING AT BASE BUILDING MAIN SUPPLY AND RETURN SHAFTS AND PROVIDE STATIC PRESSURE READINGS DOWNSTREAM AND UPSTREAM OF ALL REHEAT/HEAT COILS AND PRV.

K. THE FINAL REPORT AFTER COMFORT BALANCE IS PERFORMED SHALL BE PROVIDED TO THE BUILDING MANAGER.

L. PRE-CONSTRUCTION AIR TESTING:

a. MEASURE PRESSURE, TEMPERATURE, AND VOLUME OF AIR FROM EXISTING BASE BUILDING RETURN, SUPPLY, AND EXHAUST AIR SYSTEMS SERVING THE SCOPE OF WORK AREA BEFORE STARTING WORK.
b. MEASURE THE SUPPLY AND RETURN WATER PRESSURE FROM EXISTING BASE BUILDING CONDENSER WATER SYSTEMS SERVING THE SCOPE OF WORK AREA BEFORE STARTING WORK.
c. COORDINATE WITH BUILDING MANAGEMENT TO ENSURE SYSTEMS ARE OPERATING AT DESIGN CONDITIONS WHEN MEASUREMENTS ARE TAKEN.
d. SUBMIT REPORT TO ENGINEER IMMEDIATELY AFTER COMPLETION OF TEST.

3.06 ELECTRICAL WORK

A. ELECTRICAL CONTRACTOR SHALL BE RESPONSIBLE FOR POWER WIRING UNDER A SEPARATE DIVISION OF CONTRACT WORK. AUTOMATIC TEMPERATURE, SAFETY AND INTERLOCKING CONTROLS FOR MOTORS, MOTOR STARTERS AND OTHER ELECTRICAL APPARATUS AND DEVICES SHALL BE PROVIDED BY THE HVAC CONTRACTOR. CONTROL WIRING SHALL INCLUDE BUT NOT LIMITED TO ALL 12, 24, AND 120 VOLT WIRING.

B. THE MECHANICAL CONTRACTOR SHALL PREPARE AND SUBMIT FOR APPROVAL TERMINAL POINT TO TERMINAL POINT, COMPLETELY COORDINATED AND INTEGRATED WIRING DIAGRAMS FOR ALL WIRING REQUIRING FIELD INSTALLATION BY THE ELECTRICAL CONTRACTOR.

C. SPECIFIC WIRING DIAGRAMS OF FACTORY INSTALLED EQUIPMENT WIRING SHALL ALSO BE SUBMITTED FOR APPROVAL AND FURNISHED TO THE ELECTRICAL CONTRACTOR FOR HIS INSTALLATION REQUIREMENTS AND OTHER USES.

D. HVAC CONTRACTOR SHALL MAINTAIN ALL EXISTING CONTROL CONNECTIONS FOR STARTERS TO BE REUSED. CONTRACTOR SHALL COORDINATE EXISTING CONDITIONS AND PROVIDE ALL CONTACTS AND RELAYS REQUIRED FOR EXISTING STARTERS TO BE REPLACED WITH NEW.

E. HVAC CONTRACTOR SHALL COORDINATE WITH THE ELECTRICAL CONTRACTOR FOR THE INSTALLATION OF DUCT DETECTORS. DUCT DETECTOR SHALL BE FURNISHED AND WIRED BY THE ELECTRICAL CONTRACTOR AND MOUNTED BY THE HVAC CONTRACTOR.

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NO.	DATE	ISSUE
3	07/01/2026	ISSUE FOR ADD ALTERNATE
	05/22/2026	ISSUE FOR FILING
	04/03/2026	ISSUE FOR PRICING

REVISIONS

PROJECT:
SNOWFLAKE
7 TIMES SQUARE, NEW YORK, NY
FLOORS 29, 30, 31

DRAWING TITLE:
MECHANICAL SPECIFICATIONS SHEET 2 OF 2

SEAL & SIGNATURE

DATE: 05.22.2026
SCALE: 12" = 1'-0"
PROJECT NO. ----
DRAWN:
CHECKED:
DRAWING NO. M-102.00

NYC DOB NUMBER
M01397257-S1

DUCTWORK INSULATION

DUCTWORK LOCATION	FLAME SPREAD RATING	SMOKE DEVELOPED INDEX RATING	INSULATION MEAN TEMPERATURE RATING	MINIMUM INSTALLED INSULATION VALUE
CONDITIONED SPACE	25	50	75°F	R-6
UNCONDITIONED SPACE	25	50	75°F	R-8

DUCTWORK INSULATION NOTES:

1. REFER TO MECHANICAL SPECIFICATION DRAWINGS FOR ADDITIONAL INFORMATION.
2. ALL MINIMUM REQUIRED DUCTWORK INSULATION VALUES COMPLY WITH THE 2020 NYCECC.

BRANCH DUCT SIZING SCHEDULE

DUCT SIZE (INSIDE CLEAR DIMMENSIONS)	CFM RANGE
12x6, 15x5	UP TO 175
14x6, 18x5	180-245
12x8, 16x6, 20x5	250-285
18x6, 24x5	290-335
14x8, 20x6, 26x5	340-375
16x8, 24x6	380-435
14x10, 18x8	440-515
16x10, 20x8	520-585
22x8	590-645
18x10, 24x8	650-715
20x10, 26x8	720-825
22x10, 28x8	830-895
20x12, 24x10, 32x8	900-1,005

BRANCH DUCT SCHEDULE NOTE:

1. THIS SCHEDULE APPLIES TO BRANCH DUCTS THAT ARE NOT SIZED ON PLANS.
2. ALL SUPPLY DUCTWORK SHOWN TO HAVE ROUND DIMENSIONS SHALL BE SPIRAL ROUND DUCTWORK.

MINIMUM PIPE INSULATION

FLUID TEMPERATURE RANGE	CONDUCTIVITY RANGE (K-VALUE)	INSULATION MEAN TEMPERATURE RATING	NOMINAL PIPE DIAMETER (IN INCHES)				
			< 1	1 TO < 1.5	1.5 TO < 4	4 TO < 8	8+
			INSULATION THICKNESS REQUIRED (IN INCHES)				
141°F - 200°F	0.25 - 0.29	125°F	1.5	1.5	2.0	2.0	2.0
105°F - 140°F	0.21 - 0.28	100°F	1.0	1.0	1.5	1.5	1.5
40°F - 60°F	0.21 - 0.27	75°F	0.5	0.5	1.0	1.0	1.0
< 40°F	0.20 - 0.26	50°F	0.5	1.0	1.0	1.0	1.5

PIPE INSULATION NOTES:

1. REFER TO MECHANICAL SPECIFICATION DRAWINGS FOR ADDITIONAL INFORMATION.
2. ALL MINIMUM REQUIRED PIPING INSULATION THICKNESS COMPLY WITH THE 2020 NYCECC.

AIR OUTLET SCHEDULE

DESIGNATION	DESCRIPTION	NECK SIZE	FRAME SIZE	CFM RANGE	MAX NC	MANUFACTURER/ MODEL	COMMENTS
A	PLAQUE TYPE SUPPLY DIFFUSER	8"Ø	24"x24"	0-200	<20	TITUS/ OMNI	LAY-IN COORD. W/ CEILING TYPE
	PLAQUE TYPE SUPPLY DIFFUSER	10"Ø	24"x24"	201-350	<20	TITUS/ OMNI	LAY-IN COORD. W/ CEILING TYPE
	PLAQUE TYPE SUPPLY DIFFUSER	12"Ø	24"x24"	351-450	<20	TITUS/ OMNI	LAY-IN COORD. W/ CEILING TYPE
	PLAQUE TYPE SUPPLY DIFFUSER	14"Ø	24"x24"	451-600	<20	TITUS/ OMNI	LAY-IN COORD. W/ CEILING TYPE
	PLAQUE TYPE SUPPLY DIFFUSER	15"Ø	24"x24"	601-750	<20	TITUS/ OMNI	LAY-IN COORD. W/ CEILING TYPE
B	PLAQUE TYPE SUPPLY DIFFUSER	8"Ø	12"x12"	0-200	<20	TITUS/ OMNI	SURFACE MOUNT, G.C. TO CUT CEILING TILE AS REQUIRED
C	LINEAR SUPPLY DIFFUSER	(1) $1\frac{1}{2}$" SLOT		0-50	22	TITUS FL-15-HT	VERTICAL/HORIZONTAL MOUNT CONTINUOUS LINEAR
D	LINEAR SUPPLY DIFFUSER	(2) $\frac{3}{4}$" SLOT		0-37	22	TITUS ML-38	VERTICAL/HORIZONTAL MOUNT CONTINUOUS LINEAR
F	SUPPLY REGISTER	12"x6"	14"x8"	0-200	19	TITUS/ 300RL	SURFACE MOUNTED -
	SUPPLY REGISTER	14"x6"	16"x8"	201-300	20	TITUS/ 300RL	SURFACE MOUNTED -
	SUPPLY REGISTER	20"x6"	22"x8"	301-450	23	TITUS/ 300RL	SURFACE MOUNTED -
	SUPPLY REGISTER	22"x22"	24"x24"	750-1000	<25	TITUS/ 272RL	LAY-IN, DOUBLE DEFLECTION COORD. W/ CEILING TYPE
G	SUPPLY REGISTER	16"x4"	-	0-200	22	TITUS/ S300FL	SPIRAL DUCT MOUNTED
	SUPPLY REGISTER	22"x4"	-	201-300	23	TITUS/ S300FL	SPIRAL DUCT MOUNTED
	SUPPLY REGISTER	32"x4"	-	301-400	23	TITUS/ S300FL	SPIRAL DUCT MOUNTED
H	CEILING RETURN/ EXHAUST GRILLE	22"x22"	24"x24"	0-1570	<20	TITUS/ 355RL	$\frac{1}{2}$" SPACING, 35° DEFLECTION PROVIDE LIGHT SHEILD
I	TRANSFER DUCT RETURN GRILLE	PER PLANS	PER PLANS	0-1200	<20	TITUS/ 355RL	$\frac{1}{2}$" SPACING, 35° DEFLECTION
J	CEILING EXHAUST GRILLE	10"x10"	12"x12"	0-325	<20	TITUS/ 355RL	$\frac{1}{2}$" BLADE SPACING, 35° DEFLECTION
K	CEILING EXHAUST GRILLE	12"x12"	24"x24"	0-500	<20	TITUS/ 350FL	$\frac{3}{4}$" BLADE SPACING, 35° DEFLECTION

AIR OUTLET SCHEDULE NOTES (TYPICAL FOR EACH OUTLET):

1. INSTALL FOUR (4) WAY DIFFUSER UNLESS OTHERWISE NOTED. PROVIDE BLANK OFF BAFFLES FOR DIFFUSERS SHOWN TO HAVE 2-WAY AND 3-WAY PATTERNS. INCREASE NECK SIZES AS REQUIRED TO COMPENSATE FOR BLANKED-OFF AREA.
2. DIFFUSERS SHALL BE SUITABLE FOR THE TYPE OF CEILING CONSTRUCTION BEING INSTALLED IN.
3. DIFFUSERS THAT SERVE AREAS WITHOUT HUNG CEILINGS SHALL BE SUITABLE FOR DUCTWORK MOUNTING.
4. ALL ADJUSTABLE AIR OUTLET PATTERN DEFLECTORS SHALL BE FIELD ADJUSTED TO OPTIMIZE AIR DISTRIBUTION PREVENTING DRAFT CONDITIONS. CONTRACTOR SHALL PLAN FOR A SECOND COMFORT FIELD ADJUSTMENT PER OWNER/ENGINEER DIRECTION.
5. PROVIDE CABLE OPERATED DAMPERS FOR LINEAR DIFFUSERS AND DIFFUSERS IN INACCESSIBLE CEILINGS.
6. NON-ACTIVE LENGTHS OF LINEAR DIFFUSERS TO BE USED AS RETURN. PROVIDE LIGHT SHIELDS.
7. FINISHES SHALL BE AS SPECIFIED BY THE ARCHITECT.
8. SEE EQUIPMENT SPECIFICATIONS FOR ADDITIONAL INFORMATION.

EXHAUST FAN SCHEDULE

DESIGNATION	SERVICE	CFM	STATIC PRESSURE (IN W.G.)	ELECTRICAL V/Ø/Hz	MOTOR (HP)	MCA	MOCP	DRIVE (BELT OR DIRECT)	VFD	CONTROL (SWITCH OR THERMOSTAT)	WEIGHT	MANUFACTURER/ MODEL	COMMENTS
TF-31-1	F&B PANTRY 31.61	100	0.25	115/60/1	1/15	1.6	15	DIRECT	NO	SWITCH	27	GREENHECK/ SQ-70-VG	-
EF-30-1	DISH RM 30.13	500	0.4	115/60/1	1/10	4.75	15	DIRECT	NO	SWITCH	49	GREENHECK/ CSP-A1050-VG	-

EXHAUST FAN SCHEDULE NOTES (TYPICAL FOR EACH FAN):

1. PROVIDE FACTORY SUPPLIED VARIABLE SPEED SWITCH MOUNTED TO FAN CASING.
2. PROVIDE INTEGRAL DISCONNECT SWITCH.
3. SEE EQUIPMENT SPECIFICATIONS FOR ADDITIONAL INFORMATION.

FAN POWERED BOX SCHEDULE

DESIGNATION	BOX TYPE	FAN DATA						BOX SIZE			PRIMARY AIRFLOW RANGE (CFM)			MINIMUM STATIC PRESS. PS (IN. W.G.)	RADIATED SOUND LEVEL AT MAX CFM & PS=3" (NC)	DISCHARGE SOUND LEVEL AT MAX CFM & PS=3" (NC)	HOT WATER HEATING COIL								MANUFACTURER/ MODEL	TYPE OF CONTROLS	REMARKS
		CFM	ESP (IN W.G.)	V/Ø/Hz	HP	MCA	MOP	UNIT	INLET (INCH)	OUTLET (INCH)	MIN	MAX	HEATING				CAPACITY (MBH)	NO. OF ROWS	GPM	EWT (°F)	LWT (°F)	EAT (°F)	LAT (°F)	WATER △P (FT. HD)			
FPB-B	SERIES	325/500	0.27	277/1/60	1/3	3.2	15	B	6" / 8" Ø	20.5x 12.5"	100/150	325/500	100/150	0.04 / 0.06	13 / 20	11 / 23	-	-	-	-	-	-	-	-	TITUS/ DTFS-F	DDC	-
FPB-HW-B	SERIES	500	0.28	277/1/60	1/3	3.2	15	B	8" Ø	20.5x 12.5"	150	500	150	0.06	20	23	10.4	1	0.7	160	130	68	87	0.1	TITUS/ DTFS-F	DDC	-
FPB-C	SERIES	900	0.32	277/1/60	1/3	3.2	15	C	10" Ø	20.5x 12.5"	270	900	270	0.11	20	23	-	-	-	-	-	-	-	-	TITUS/ DTFS-F	DDC	-
FPB-HW-C	SERIES	900	0.32	277/1/60	1/3	3.2	15	C	10" Ø	20.5x 12.5"	270	900	270	0.11	20	23	12.2	1	0.7	160	125	68	80	0.1	TITUS/ DTFS-F	DDC	-
FPB-D	SERIES	1500	0.31	277/1/60	1/2	5.1	15	D	12" Ø	25x 17.5"	450	1500	450	0.08	24	27	-	-	-	-	-	-	-	-	TITUS/ DTFS-F	DDC	-
FPB-HW-D	SERIES	1500	0.31	277/1/60	1/2	5.1	15	D	12" Ø	25x 17.5"	450	1500	450	0.08	24	27	23.3	1	1.2	160	119	68	82	0.34	TITUS/ DTFS-F	DDC	-
FPB-E	SERIES	1700	0.33	277/1/60	3/4	6.9	15	E	14" Ø	25x 17.5"	510	1700	510	0.06	29	28	-	-	-	-	-	-	-	-	TITUS/ DTFS-F	DDC	-
FPB-HW-E	SERIES	1700	0.33	277/1/60	3/4	6.9	15	E	14" Ø	25x 17.5"	510	1700	510	0.06	29	28	24.9	1	1.2	160	119	68	81	0.36	TITUS/ DTFS-F	DDC	-

FAN COIL UNIT SCHEDULE:

1. ALL PERFORMANCE BASED ON TESTS CONDUCTED IN ACCORDANCE WITH ASHRAE 130-2009 AND AHRI 880-2008.
2. ALL NC LEVELS DETERMINED USING AHRI 885-2008 APPENDIX E.
3. THE MINIMUM SUPPLY CIRCUIT AMPACITY (MCA) AND MAXIMUM OVER-CURRENT PROTECTION (MOP) RATINGS WERE CALCULATED IN ACCORDANCE WITH UL STANDARDS BASED ON MOTOR FULL LOAD CURRENT RATINGS
4. CONTROLS SHALL BE PROVIDED BY OTHERS, SHIPPED TO FACTORY AND FACTORY MOUNTED.
5. FAN POWER BOX TO BE PROVIDED WITH ECM MOTOR AND MULTI-POINT CENTER AVERAGING INLET VELOCITY SENSOR CAPABLE OF OPERATING AT VELOCITIES AS LOW AS 250 FPM
6. OPTIONAL REMOTE PWM ALLOWS FOR A 0-10V SIGNAL TO MODULATE FAN RPM TO MIN/MAX, IF BMS NEEDS TO STOP THE FAN OPTIONAL CONTROL SEQUENCE IS AVAILABLE WITH 0-2V SINGAL ACTING AS A STOP SIGNAL AND 2.,1-10V SIGNAL AS A FAN MIN/MAX SETTING. REVIEW SPECIFICATIONS FOR EXACT LOCATIONS AND CONTROLS NECESSARY.
7. PROVIDE FACTORY MOUNTED BRACKETS AND TOGGLE DISCONNECT SWITCH.
8. PROVIDE FILTER RACK WITH TWO (2) FACTORY SUPPLIED FILTERS FOR RETURN INLET (ONE (1) FOR CONSTRUCTION USE AND THROWAWAY)
9. CONTRACTOR TO SUBMIT INDIVIDUAL BOX SCHEDULE FOR REVIEW DURING THE SHOP DRAWING PROCESS. ACCESS SIDE SHALL BE INDICATED AND COORDINATED WITH THE DESIGN AND SHEETMETAL SHOP DRAWINGS
10. PRIMARY AIR TEMPERATURE IS BASED ON 49°F, RETURN AIR BASED ON 70°F.
11. PROVIDE 277 VOLT TO 24 VOLT TRANSFORMER.
12. U.O.N. EACH BOX SHALL HAVE A MINIMUM AIRFLOW SETPOINT OF 25% ON PRIMARY AIR INLET.
13. UNITS SHALL BE CONSTRUCTED OF A MINIMUM 20-GAUGE GALVANIZED STEEL.
14. SEE SPECS FOR MORE INFORMATION.

NOTE: ALL FAN POWERED BOXES LABEL FPB-HW ON SHEETS SHALL BE PROVIDED WITH HOT WATER HEATING COIL.

WATER COOLED COMPUTER ROOM AIR CONDITIONING UNIT SCHEDULE

DESIGNATION	AREA SERVED	CAPACITY			SPACE CONDITION			AIR FLOW				INFRARED HUMIDIFIER		CONDENSER SECTION					ELECTRICAL V/Ø/Hz	UNIT FLA	UNIT MCA	UNIT MOP	OPERATING WEIGHT (LBS) (EVAP / COND)	MANUFACTURER/ MODEL (EVAP / COND)	COMMENTS
		NOMIAL TONS	TOTAL (MBH)	SENSIBLE (MBH)	DB (°F)	WB (°F)	% RH	CFM	ESP	VFD	LAT (DB/WB °F)	LBS/HR	FLA	DESIGNATION	GPM	MAX PD (FT)	EWT (°F)	LWT (°F)							
CRAC-29-1	IDF-29.35	3.0	37.5	37.5	75	62.5	50	1500	0.5	NO	57.0/42.1	5	8.2	WCCU-29-1	9.0	10.7	85	95	460/3/60	16.5	20.6	25	205 / 150	ABOVE AIR/ WCH-011A / CCW-011B	WALL MOUNTED UNIT
CRAC-30-1	IDF-30.29	3.0	37.5	37.5	75	62.5	50	1500	0.5	NO	57.0/42.1	5	8.2	WCCU-30-1	9.0	10.7	85	95	460/3/60	16.5	20.6	25	205 / 150	ABOVE AIR/ WCH-011A / CCW-011B	WALL MOUNTED UNIT
CRAC-31-1	IDF-31.44	3.0	37.5	37.5	75	62.5	50	1500	0.5	NO	57.0/42.1	5	8.2	WCCU-31-1	9.0	10.7	85	95	460/3/60	16.5	20.6	25	205 / 150	ABOVE AIR/ WCH-011A / CCW-011B	WALL MOUNTED UNIT

WATER COOLED COMPUTER ROOM AIR CONDITIONING UNIT SCHEDULE NOTES (TYPICAL FOR EACH UNIT)

1. PROVIDE SPRING TYPE VIBRATION ISOLATION HANGERS FOR EACH UNIT.
2. PROVIDE MC-SERIES ADVANCED MICROPROCESSOR W/ ALARMS & BACNET COMPATIBILITY.
3. PROVIDE FACTORY SUPPLIED COMBINATION STARTER AND DISCONNECT SWITCH.
4. PROVIDE AUXILIARY STAINLESS STEEL DRAIN PAN WITH LEAK DETECTION.
5. PROVIDE LOW ENTERING WATER TEMP KIT.
6. UNITS TO BE RATED FOR SYSTEM PRESSURE.
7. PROVIDE HOT GAS BYPASS.
8. FACTORY INSTALLED ACCESSORIES: CONDENSATE PUMP, RECEIVER, COMPRESSOR SOUND JACKET, FIRESTAT, SMOKE DETECTOR, HUMIDIFIER.
9. SEE EQUIPMENT SPECIFICATIONS FOR ADDITIONAL INFORMATION.

PIPING FITTING SCHEDULE

PIPE MATERIAL	PIPE SIZE	JOINT TYPE	FITTING MATERIAL	FITTING CLASS
COPPER TUBING HARD DRAWN	4" & SMALLER	BRAZING	WROUGHT COPPER	450 PSIG AT 100°F TO 200°F, 150 PSIG AT 250°F
RED BRASS	ALL SIZES	THREADED	CAST BRONZE	125 AND 250 PSIG

PIPING FITTING NOTES:

1. REFER TO SPECIFICATIONS FOR MORE INFORMATION.

AIR PURIFICATION SYSTEM SCHEDULE

DESIGNATION	SERVING	SUPPLY AIR FLOW CFM	FAN			STAGES OF FILTRATION				ELECTRICAL		UNIT WEIGHT (LBS)	UNIT DIMENSIONS (LxWxH)	MANUFACTURER/ MODEL	COMMENTS
			AMP	RPM	HP	1	2	3	4	V / Ø / Hz	FLA				
BAS-30-1,2	FOOD PREP/ SERVING PT	500	1.5	1725	1/8	1" PREFILTER	(2) 2" CARBON MEDIA	4" PRIMARY PARTICULATE FILTER	(2) BI-POLAR IONIZATION MODULES	115/1/60	5.0	90	48x24x12	BIOCLIMATIC/ MC 880	- -

BIOCLIMATIC SYSTEM SCHEDULE NOTES:

1. UNIT SHALL BE A SELF-CONTAINED FLUSH MOUNTED SYSTEM FOR INSTALLATION INTO "T" GRID CEILING, 48"X24" MODULE1"
2. ACOUSTICAL FOAM INSULATION
3. SYSTEM SHALL DELIVER SPECIFIED CFM WITH ALL FILTERS INSTALLED IN UNIT
4. ALL METAL TO METAL SURFACES SEALED WITH SILICONE SEALANT
5. SOUND LEVEL NOT TO EXCEED 53.5 DB @ 4.5 FT
6. FAN SHALL BE DWDI FORWARD CURVE WHEEL, TWIN UNITS
7. TWO (2) AIR DIFFERENTIAL PRESSURE SWITCHES FOR PREFILTER AND PRIMARY FILTER CHANGE
8. DIRTY FILTER LIGHT
9. EQUIPMENT SHALL BE ETL LISTED
10. UNIT MOUNTED CONTROLS FOR STANDALONE AND ON/OFF OPERATION
11. UNIT MANUFACTURER TO PROVIDE ONE FACTORY STARTUP AND ONE ANNUAL FOLLOW UP INSPECTION WITHIN 1 YEAR OF STARTUP
12. UNIT INSTALLATION AND ALL WIRING IS BY OTHERS
13. MEDIA SHIPPED SEPARATELY FOR FIELD INSTALLATION BY OTHERS
14. VIBRATION ISOLATORS AND MOUNTING HARDWARE BY OTHERS
15. MINIMUM 36" CLEARANCE REQUIRED FOR FILTER CHANGEOUT
16. UNITS NOT TO EXCEED 48"LX24"WX14"H DIMENSIONS
17. ONE (1) SPARE MEDIA FILTER SET PER UNIT SHALL BE PROVIDED.

PUMP SCHEDULE

DESIGNATION	SERVICE	TYPE	GPM	TDH (FT)	RPM	NPSH	PUMP EFFICIENCY	BHP	MOTOR HP	VFD	ELECTRICAL V/Ø/Hz	WEIGHT (LBS.)	MANUFACTURER/ MODEL	COMMENTS
CWP-29-1,2	CONDENSER WATER	INLINE MULTISTAGE	10	35	2,845	3.85	84	-	0.5	YES	208/1/60	73	GRUNDFOS/ CRE3-DOL2VFD	TWO PUMPS PRIMARY-STANDBY
CWP-30-1,2	CONDENSER WATER	INLINE MULTISTAGE	10	35	2,845	3.85	84	-	0.5	YES	208/1/60	73	GRUNDFOS/ CRE3-DOL2VFD	TWO PUMPS PRIMARY-STANDBY
CWP-31-1,2	CONDENSER WATER	INLINE MULTISTAGE	10	35	2,845	3.85	84	-	0.5	YES	208/1/60	73	GRUNDFOS/ CRE3-DOL2VFD	TWO PUMPS PRIMARY-STANDBY

PUMP SCHEDULE NOTES (TYPICAL FOR EACH PUMP):

1. PROVIDE SPRING TYPE VIBRATION AT PUMP.
2. PROVIDE COMBINATION DISCONNECT SWITCH AND STARTER AS REQUIRED.
3. PROVIDE DRIP PAN UNDER IN-LINE PUMPS WITH LEAK DETECTOR.
4. PUMPS TO BE RATED FOR 363 PSI.
5. MECHANICAL CONTRACTOR TO VERIFY SUPPLY AND RETURN DIFFERENTIAL PRESSURE OF EXISTING RISERS IN FIELD AT POINT OF CONNECTION AND SUBMIT PRESSURE READINGS WITH PUMP SHOP SUBMITTALS FOR REVIEW AND RECORD.
6. PROVIDE PERMANENT MAGNET ECM MOTOR (IE5 EFFICIENCY) WITH INTEGRAL VFD.
7. MECHANICAL CONTRACTOR TO PROVIDE NON OVERLOADING PUMP CURVE FOR REVIEW DURING THE SHOP DRAWING PROCESS. FOR VFD APPLICATIONS PROVIDE PARALLEL PUMP SPEED CURVES IN 10% INCREMENTS.
8. INSTALLATION SHALL BE IN STRICT ACCORDANCE WITH THE MANUFACTURERS INSTRUCTIONS.
9. SEE EQUIPMENT SPECIFICATIONS FOR ADDITIONAL INFORMATION.

DEHUMIDIFICATION UNIT SCHEDULE - ADD ALT

DESIGNATION	AREA SERVED	CAPACITY		SPACE CONDITION			FILTER	EFFICIENCY (PINTS/KWH)	BTU (TOTAL)	ELECTRICAL V/Ø/Hz	UNIT MCA	UNIT MOP	OPERATING WEIGHT (LBS)	MANUFACTURER/ MODEL	COMMENTS
		TOTAL (MBH)	WATER REMOVAL (@ 80°F / 60RH%)	MIN TEMP (°F)	MAX TEMP (°F)	% RH									
DU-29-1	IDF-29.35	37.5	200	56	95	50	MERV-13	5.7	12,800	120/1/60	15	25	130	QUEST/ 205 DUAL	ADD ALT
DU-30-1	IDF-30.29	37.5	200	56	95	50	MERV-13	5.7	12,800	120/1/60	15	25	130	QUEST/ 205 DUAL	ADD ALT
DU-31-1	IDF-31.44	37.5	200	56	95	50	MERV-13	5.7	12,800	120/1/60	15	25	130	QUEST/ 205 DUAL	ADD ALT

DEHUMIDIFICATION UNIT SCHEDULE NOTES (TYPICAL FOR EACH UNIT)

1. PROVIDE SPRING TYPE VIBRATION ISOLATION HANGERS FOR EACH UNIT.
2. PROVIDE AUXILIARY STAINLESS STEEL DRAIN PAN WITH LEAK DETECTION.
3. TO BE INSTALLED ACCESSORIES: CONDENSATE PUMP

PIPING MATERIAL SCHEDULE

SERVICE	SIZE	MATERIAL	WEIGHT	STANDARD	JOINT TYPE
COLD CONDENSATE DRAINS, MISCELLANEOUS DRAINS AND OVERFLOWS	2" & BELOW	HARD COPPER	TYPE 'L'	ASTM A88	BRAZE
HOT WATER TO 220°F	4" & BELOW	HARD COPPER	TYPE 'K'	ASTM B88	BRAZE
CONDENSER WATER	4" & BELOW	HARD COPPER	TYPE 'K'	ASTM B88	BRAZE
AUTOMATIC AIR VENT	ALL	HARD COPPER	TYPE 'K'	ASTM B88	BRAZE
COLD WATER MAKEUP AND FILL	4" & BELOW	HARD COPPER	TYPE 'L'	ASTM B88	BRAZE
PIPING TO MISC. GAUGES	ALL	RED BRASS	STANDARD	ASTM B43	THREADED

PIPING MATERIAL NOTES:

1. REFER TO SPECIFICATIONS FOR MORE INFORMATION.

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Bellevue, WA 98004

NO.	DATE	ISSUE
3	07/01/2026	ISSUE FOR ADD ALTERNATE
	05/22/2026	ISSUE FOR FILING
1	04/22/2026	ADDENDUM #1
	04/03/2026	ISSUE FOR PRICING

REVISIONS

PROJECT:

SNOWFLAKE
7 TIMES SQUARE, NEW YORK, NY
FLOORS 29, 30, 31

DRAWING TITLE:

MECHANICAL SCHEDULES

SEAL & SIGNATURE

DATE: 05.22.2026
SCALE: N.T.S.
PROJECT NO. ----
DRAWN:
CHECKED:
DRAWING NO. **M-401.00**

NYC DOB NUMBER

M01397257-S1

CMTA
A LEGENCE Company
Formerly AMA Group

Worldwide Plaza | 825 Eighth Avenue, FL18
New York, NY 10019 | 212.944.7722

CMTA Project No.:	B069-02-039	**Control No.:**	**H-02.00**
CMTA Project Name:	Boston Properties @ 7 Times Square - Snowflake Floors 29-31	**Page:**	1 **of** 1

☐ **NO EXCEPTIONS TAKEN** No further review of submittal required.	☐ **REVISE AND RESUBMIT** Revise as noted and resubmit for further review.
☒ **MAKE CORRECTIONS AS NOTED** Incorporate corrections in work; resubmission is not required. Where corrections cannot be complied with, revise the exceptions and resubmit.	☐ **EXAMINED** Submittal received, formal review by others and/or formal review by this office not required. Submittal has been examined for information only.

Corrections or comments made on the shop drawings during this review do not relieve contractor from compliance with requirements of the drawings and specifications. This check is only for review of general conformance with the design concept of the project and general compliance with the information given in the contract documents. The contractor is responsible for: confirming and correlating all quantities and dimensions; selecting fabrication processes and techniques of construction; coordinating his work with that of all other trades; and performing his work in a safe and satisfactory manner.

Date:	6/15/2026	**Engineer**:	Mike Sisti

SUBMITTAL DESCRIPTION: Pump Cuts

COMMENTS:

CMTA Comments:

1. Provide vibration isolation for pumps and piping within 50 feet or 50 pipe diameters, whichever is greater. See specifications for details.

2. Provide valve and trim as per the details on the design documents.

3. Pumps shall meet all pressure and flow characteristics as per the design documents.

4. Contractor to provide the differential pressure at the tie in to the risers in full flow condition, as per the design documents, and provide to the manufacturer to ensure the proper head is obtained to ensure full flow throughout the system.